

Competition by Design:

Why the United States Must Deconsolidate Its Defense Industrial Base

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Executive Summary

In every one of the twenty-four Taiwan-invasion wargames the Center for Strategic and International Studies (CSIS) ran in January 2023, the United States exhausted its entire global inventory of Long Range Anti-Ship Missiles (LRASM), roughly 450 rounds, within the first week of the fight. The production line supplying LRASM built just 88 missiles that year and needed roughly two years to expand.¹ The same failure recurs in artillery. The Army's 155mm artillery-shell ramp, launched from a baseline of 14,000 rounds a month in 2022, missed a series of its own interim targets and still produced only 40,000 rounds a month by June 2025; its ultimate goal of 100,000 rounds a month has slipped from October 2025 to mid-2026.² Against Russian jamming in Ukraine, the Excalibur guided artillery shell's hit rate fell from 55 percent in January 2023 to 6 percent that August, a collapse steep enough that Ukrainian units stopped firing it.³

The failures are structural. The Pentagon's 1993 "Last Supper" collapsed 51 aerospace and defense prime contractors into 5 within a decade, and the Department of Defense's own 2022 accounting shows individual weapons categories narrowing the same way underneath them.⁴

¹Mark F. Cancian, Matthew Cancian, and Eric Heginbotham. *The First Battle of the Next War: Wargaming a Chinese Invasion of Taiwan*. Center for Strategic and International Studies, Jan. 9, 2023. URL: https://csis-website-prod.s3.amazonaws.com/s3fs-public/publication/230109_Cancian_FirstBattle_NextWar.pdf (visited on 08/21/2026).

²Jen Judson. "Army Races to Widen the Bottlenecks of Artillery Shell Production." In: *Defense News* (Oct. 14, 2024). URL: <https://www.defensenews.com/land/2024/10/14/army-races-to-widen-the-bottlenecks-of-artillery-shell-production/> (visited on 08/21/2026); Breaking Defense. "Army Hitting Stride with 155mm Production, but Key General Worries over What's Needed Next." In: *Breaking Defense* (Oct. 17, 2025). URL: <https://breakingdefense.com/2025/10/army-hitting-stride-with-155mm-production-but-key-general-worries-over-whats-needed-next/> (visited on 08/21/2026).

³Leo Chiu and Christopher Stewart. "Off Target: Are Kyiv's GPS-Aided Weapons Susceptible to Kremlin EW Jamming?" In: *Kyiv Post* (May 29, 2024). URL: <https://www.kyivpost.com/analysis/33448> (visited on 08/21/2026).

⁴WBUR. *The Last Supper: How a 1993 Pentagon Dinner Reshaped the Defense Industry*. WBUR, On Point. Mar. 1, 2023. URL: <https://www.wbur.org/onpoint/2023/03/01/the-last-supper-how-a-1993-pentagon-dinner-reshaped-the-defense-industry> (visited on 08/21/2026); U.S. Department of Defense. *State of Competition within the Defense Industrial Base*. U.S. Department of Defense, Feb. 2022. URL: <https://media.defense.gov/2022/Feb/15/2002939087/-1/-1/1/STATE-OF-COMPETITION-WITHIN-THE->

Solid rocket motors, intercontinental ballistic missiles (ICBMs), most air-defense interceptors, submarines, and towed artillery are sole- or dual-source segments today, the direct legacy of that collapse.⁵

The 2025–26 reform wave moved fast. The fiscal 2026 National Defense Authorization Act (NDAA) and the Department of War’s Warfighting Acquisition System led it: the NDAA alone amended or repealed dozens of prior acquisition statutes and codified a new tier of portfolio-level program management.⁶ But every one of those levers is a process lever. None mandates dual-sourcing, none authorizes divestiture, and none requires a chokepoint buy to be split across suppliers by statute; the Pentagon’s own merger-review tool, Section 857, only notifies antitrust regulators and cannot block a deal.⁷

The 2025–26 acquisition reforms fixed process; they enacted no structural remedy. Competition must be designed back into the base where the evidence says it pays: dual-sourcing where volume and learning-curve conditions clear RAND’s breakeven thresholds, entry protection where new suppliers are already arriving, and divestiture only where both fail. This paper recommends:

1. Congress should mandate dual-sourcing in every chokepoint segment where volume and learning-curve conditions clear RAND’s breakeven thresholds.
2. Congress should require portfolio buys in chokepoint segments to be split across qualified suppliers by statute, not left to program-by-program discretion.
3. The Department of Justice and Federal Trade Commission should reserve divestiture for chokepoints where dual-sourcing cannot be seeded, such as solid rocket motors and specialty energetics.
4. Congress should convert the Pentagon’s Section 857 merger notification into real authority to block further prime-contractor consolidation.
5. The Department of Defense should fund and fast-track new suppliers already qualifying inside chokepoint segments, the cheaper alternative to breakup.

DEFENSE-INDUSTRIAL-BASE.PDF (visited on 08/21/2026).

⁵Seth G. Jones et al. *Empty Bins in a Wartime Environment: The Challenge to the U.S. Defense Industrial Base*. Center for Strategic and International Studies, Jan. 19, 2023. URL: https://csis-website-prod.s3.amazonaws.com/s3fs-public/2023-01/230119_Jones_Empty_Bins.pdf (visited on 08/21/2026).

⁶Holland & Knight. *FY 2026 National Defense Authorization Act*. Holland & Knight. Dec. 22, 2025. URL: <https://www.hklaw.com/en/insights/publications/2025/12/fy-2026-national-defense-authorization-act> (visited on 08/21/2026); U.S. Department of War. *Transforming the Defense Acquisition System into the Warfighting Acquisition System to Accelerate Fielding of Urgently Needed Capabilities to Our Warriors*. U.S. Department of War, Nov. 7, 2025. URL: <https://media.defense.gov/2025/Nov/10/2003819439/-1/-1/1/TRANSFORMING-THE-DEFENSE-ACQUISITION-SYSTEM-INTO-THE-WARFIGHTING-ACQUISITION-SYSTEM-TO-ACCELERATE-FIELDING-OF-URGENTLY-NEEDED-CAPABILITIES-TO-OUR-WARRIORS.PDF> (visited on 08/21/2026).

⁷Greenberg Traurig. *New Guidance for Defense Sector M&A Review*. Greenberg Traurig. Feb. 26, 2026. URL: <https://www.gtlaw.com/en/insights/2026/2/new-guidance-for-defense-sector-m-and-a-review> (visited on 08/21/2026).

1 Introduction

In January 2023 the Center for Strategic and International Studies (CSIS) ran twenty-four iterations of a wargamed Chinese invasion of Taiwan. In every one of them, the United States exhausted its entire global inventory of Long Range Anti-Ship Missiles (LRASM), roughly 450 rounds, within the first week of the fight. Across those same iterations the broader munitions bill ran past 5,000 long-range missiles in three weeks, a rate the compressed production line supplying LRASM was never built to sustain: it turned out just 88 rounds that year, against a roughly two-year lead time to expand.⁸ The math never closed.

The wargame’s own authors traced the shortfall to a portfolio problem, not a technology one: LRASM, the missile purpose-built to sink ships, sat undersupplied on the same production line as the Joint Air-to-Surface Standoff Missile (JASSM), the general-purpose missile the United States happened to procure in bulk instead. Three years later the gap has not closed; a January 2026 Heritage Foundation assessment found the global LRASM inventory still short of what analysts say a Taiwan contingency would require.⁹ A shortfall this severe, repeated in wargame after wargame, is what a supplier base narrowed to a handful of firms produces when demand spikes.

This is not new: in July 1993 Deputy Secretary of Defense William Perry told the assembled chief executives of the American aerospace and defense industry that the government could sustain only one to three firms in each of sixteen weapons categories, and that Washington would not intervene. “We expect defense companies to go out of business,” Perry said afterward. “We will stand by and watch it happen.”¹⁰ Fifty-one prime contractors became five within a decade, and the Department of Defense’s own 2022 accounting shows individual weapons categories narrowing the same way underneath them: tactical missiles from 13 suppliers to 3, fixed-wing aircraft from 8 to 3, satellites from 8 to 4.¹¹ The chokepoints today’s wargames expose are the direct descendants of that consolidation.

Congress and the Pentagon spent 2025 and 2026 rewriting acquisition procedure. The fiscal 2026 National Defense Authorization Act (NDAA) amended or repealed dozens of prior acquisition statutes and created a new tier of Portfolio Acquisition Executives; the Department of War’s Warfighting Acquisition System reorganized program offices around the same portfolio logic days later.¹² Congress is not simply deferential here: appropriators rejected part of the Pentagon’s own budget-flexibility request as “premature” in January 2026.¹³ But no version

⁸Cancian, Cancian, and Heginbotham, *The First Battle of the Next War: Wargaming a Chinese Invasion of Taiwan*, op. cit.; Jones et al., *Empty Bins in a Wartime Environment: The Challenge to the U.S. Defense Industrial Base*, op. cit.

⁹The Heritage Foundation. *TIDALWAVE, Chapter 6: Assessing the U.S. Indo-Pacific Munitions System*. The Heritage Foundation, Jan. 20, 2026. URL: <https://www.heritage.org/tidalwave/chapters/chapter-6-assessing-the-us-indo-pacific-munitions-system> (visited on 08/21/2026).

¹⁰WBUR, *The Last Supper: How a 1993 Pentagon Dinner Reshaped the Defense Industry*, op. cit.

¹¹U.S. Department of Defense, *State of Competition within the Defense Industrial Base*, op. cit.

¹²Holland & Knight, *FY 2026 National Defense Authorization Act*, op. cit.; U.S. Department of War, *Transforming the Defense Acquisition System into the Warfighting Acquisition System to Accelerate Fielding of Urgently Needed Capabilities to Our Warriors*, op. cit.

¹³Anastasia Obis. “Congress Pushes Back on Parts of DoD’s Acquisition Reform Agenda.” In: *Federal News*

of the 2025–26 package included a dual-source mandate, a divestiture trigger, or a statutory requirement to split a chokepoint buy across suppliers. Process changed; structure did not.

This paper argues that the gap will not close on its own. The 2025–26 acquisition reforms fixed process; they enacted no structural remedy. Competition must be designed back into the base where the evidence says it pays: dual-sourcing where volume and learning-curve conditions clear RAND’s breakeven thresholds, entry protection where new suppliers are already arriving, and divestiture only where both fail.

The argument proceeds in six moves. Section 2 traces how a demand-driven consolidation in the 1990s produced today’s sole-source chokepoints, and engages the strongest revisionist accounts of that history fairly. Section 3 maps those chokepoints against the surge failures, electronic-warfare brittleness, cost-exchange asymmetries, and Chinese production capacity the last four years exposed. Section 4 tests the remedy against the historical record: America’s own Great Engine War, Ukraine’s distributed drone base, and the Soviet design-bureau precedent, including the cases where dual-sourcing failed and modern Russia’s consolidated counterexample. Section 5 assesses what the 2025–26 reform wave actually enacted against what it left undone. Section 6 lays out the five-part structural program, dual-sourcing, portfolio splits, divestiture, merger policy, and entry subsidies, that this paper’s evidence supports and prices. Section 7 answers the strongest objections, including the case that entry alone suffices, and closes.

This paper is not a neutral survey. It takes a position, names a structural fix, and asks Congress, the Department of Defense, and antitrust regulators to adopt it. That advocacy is stated here, once, before the evidence, not left for the reader to infer from Section 6’s recommendations. The next war will not wait for a paperwork fix.

2 How the Base Consolidated

The Dinner

On July 21, 1993, Secretary of Defense Les Aspin and Deputy Secretary William Perry invited roughly twenty to twenty-five chief executives of the country’s defense firms to a private dinner in the Pentagon’s executive dining room.¹⁴ Norm Augustine, then chairman and chief executive of Martin Marietta, recalled Aspin’s warning as he sat down: “in about 15 minutes, you’re going to find out. You probably aren’t going to like it.”¹⁵ Perry’s chart covered sixteen equipment categories and how many prime contractors the Pentagon believed each could still support: three categories could support three firms, six could support two, and seven could support only one.¹⁶ The Pentagon would not pick winners; Augustine paraphrased it: “the government was not in the business of redesigning companies or consolidating industries or putting people in or

Network (Jan. 20, 2026). URL: <https://federalnewsnetwork.com/congress/2026/01/congress-pushes-back-on-parts-of-dods-acquisition-reform-agenda/> (visited on 08/21/2026).

¹⁴WBUR, *The Last Supper: How a 1993 Pentagon Dinner Reshaped the Defense Industry*, op. cit.

¹⁵*Ibid.*

¹⁶*Ibid.*

out of business. That was up to us.”¹⁷

Perry made the same point in public, more bluntly: “We expect defense companies to go out of business. We will stand by and watch it happen.”¹⁸ Augustine coined the dinner’s nickname to a reporter the next morning, and it stuck: the Last Supper.¹⁹

A Rational Response to a Collapsing Budget

The dinner did not happen in a vacuum. Defense procurement spending had been falling for years before Perry’s chart, and kept falling for years after. The Government Accountability Office (GAO) found Department of Defense (DoD) procurement outlays fell 52 percent in real terms between FY1987 and FY1995, the lowest level since 1946, the fourth major postwar drawdown and, from 1985 to 1995, its longest consistent decline.²⁰

CSIS’s own accounting shows Army procurement funding fell even further, from \$33.90 billion in FY1985 to \$10.11 billion in FY1998, a 70 percent real decline.²¹ The broader defense enterprise felt the same contraction: the overall budget fell more than 15 percent from FY1991 to FY1996,²² and defense-related employment dropped roughly 39 percent from 1989 to 1997.²³ The share of the national workforce in manufacturing fell from 18 percent in 1980 to 7 percent shortly after the Last Supper, in Augustine’s words.²⁴

This longer cycle is visible in Figure 1: real procurement funding fell from a FY1985 peak of \$218.6 billion to a FY1997 trough of \$77.8 billion, and did not approach that peak again until the FY2026 enacted budget of \$214.7 billion, still short of 1985 and further below the FY2008 wartime high. Perry was not inventing scarcity. He was responding to it.

Fifty-One to Five

By 1997 the wave Perry’s dinner helped accelerate had run its course. DoD’s own 2022 accounting counts roughly forty mergers between 1992 and 1997 collapsing fifty-one prime contractors into five: Lockheed Martin, Raytheon, General Dynamics, Northrop Grumman, and Boeing, a reduction of more than 90 percent.²⁵ A prime contractor disappeared roughly every six weeks. The pace was not accidental: that summer, when Justice Department antitrust staff

¹⁷Ibid.

¹⁸Ibid.

¹⁹Ibid.

²⁰U.S. Government Accountability Office. *Defense Industry: Trends in DOD Spending, Industrial Productivity, and Competition*. GAO/PEMD-97-3. U.S. Government Accountability Office, Jan. 31, 1997. URL: <https://www.gao.gov/assets/pemd-97-3.pdf> (visited on 08/21/2026).

²¹Rhys McCormick. *The Army Modernization Challenge*. Center for Strategic and International Studies, Defense360, Mar. 28, 2016. URL: <https://defense360.csis.org/army-modernization-challenge-historical-perspective-2/> (visited on 08/21/2026).

²²WBUR, *The Last Supper: How a 1993 Pentagon Dinner Reshaped the Defense Industry*, op. cit.

²³U.S. Government Accountability Office, *Defense Industry: Trends in DOD Spending, Industrial Productivity, and Competition*, op. cit.

²⁴WBUR, *The Last Supper: How a 1993 Pentagon Dinner Reshaped the Defense Industry*, op. cit.

²⁵U.S. Department of Defense, *State of Competition within the Defense Industrial Base*, op. cit.

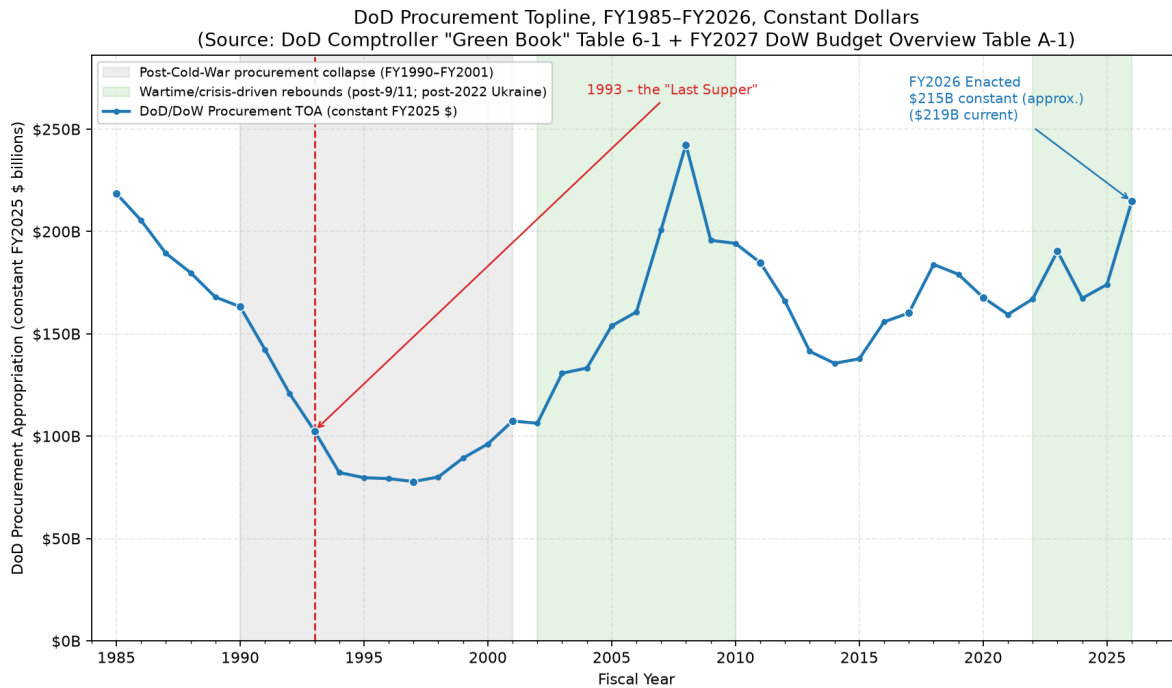


Figure 1: Procurement funding has not regained its FY1985 peak; the July 1993 Last Supper dinner sits partway down that decline.

Source: U.S. Department of Defense (Department of War) Comptroller, *National Defense Budget Estimates for FY2025* ("Green Book"), Table 6-1; Office of the Under Secretary of War (Comptroller), *FY2027 Department of War Budget Overview*, Table A-1.

balked at Martin Marietta's purchase of General Electric's aerospace business, Augustine called Perry, and the resistance eased.²⁶

Lockheed and Martin Marietta merged outright in March 1995 to form Lockheed Martin.²⁷ Boeing agreed to buy McDonnell Douglas for \$14 billion in December 1996.²⁸ Only when Lockheed Martin moved to acquire Northrop Grumman for \$11.6 billion in July 1997 did Washington reverse course: Attorney General Janet Reno's Justice Department moved in March 1998 to block the deal, which died.²⁹

The wave's result, category by category, appears in Table 1.³⁰

The same report found that 90 percent of missiles came from three sources, and that the top five primes alone collected more than 25 percent of all DoD procurement dollars in FY2021,

²⁶WBUR, *The Last Supper: How a 1993 Pentagon Dinner Reshaped the Defense Industry*, op. cit.

²⁷John Mintz. "How a Dinner Led to a Feeding Frenzy." In: *The Washington Post* (July 3, 1997). URL: <http://web.archive.org/web/20240512012154/https://www.washingtonpost.com/archive/business/1997/07/04/how-a-dinner-led-to-a-feeding-frenzy/13961ba2-5908-4992-8335-c3c087cdebc6/> (visited on 08/21/2026).

²⁸WBUR, *The Last Supper: How a 1993 Pentagon Dinner Reshaped the Defense Industry*, op. cit.

²⁹Mintz, "How a Dinner Led to a Feeding Frenzy," op. cit.

³⁰U.S. Department of Defense, *State of Competition within the Defense Industrial Base*, op. cit.

Table 1: U.S.-based prime contractors by weapons category, 1990–2020.

Source: U.S. Department of Defense, *State of Competition within the Defense Industrial Base* (Feb. 2022), Table 1.

Weapons category	1990	1998	2020
Tactical missiles	13	3	3
Fixed-wing aircraft	8	3	3
Satellites	8	5	4
Surface ships	8	5	2
Tracked combat vehicles	3	2	1

“roughly the same amount that DoD expended on the next 95 defense contractors combined”.³¹ Three of these categories translate into the Herfindahl-Hirschman Index (HHI) in Figure 2. Even under a conservative equal-share assumption, tactical missiles, fixed-wing aircraft, and surface ships had all crossed the Department of Justice (DOJ) and Federal Trade Commission (FTC) “highly concentrated” threshold by the early 2000s and have not fallen back since. Surface ships are now the most concentrated of the three, at an HHI of 5,000 on two suppliers, down from eight in 1990. By the same DOJ/FTC thresholds, three categories have stayed highly concentrated for two decades.

The History Was Messier, the Diagnosis Still Holds

The concession here is causal weight only, not the concentration or its chokepoints. Margaret Mullins of the Vanderbilt Policy Accelerator presses the causal case hardest. She argues the forces that mattered ran on a longer clock than one dinner: 1980s financialization, decades of privatization, and an erosion of the government’s own organic acquisition capacity were consolidating the industry before Perry’s chart and kept doing so after it.³² She calls it a “Long Supper,” and her fix is rebuilding government capacity to manage the market as its sole buyer, not blaming one meeting.³³ Mullins does not dispute that fifty-one primes became five or that the result produced today’s chokepoints.³⁴

Her account is right about the trend line and still misses the point: the 1993 intervention and 1998 block above show government capacity was not absent, only inconsistently used.³⁵ Washington opened the door on consolidation and closed it again within five years; fifty-one companies collapsed into five regardless. Structure, not capacity, was the missing ingredient.

John Driessnack of the Naval Postgraduate School questions whether 51-to-5 is evidence of

³¹Ibid.

³²Margaret Mullins. *The Myths of the Last Supper: The Lessons of History and the Future of Defense Procurement*. Vanderbilt Policy Accelerator, Sept. 2025. URL: <https://cdn.vanderbilt.edu/vu-URL/wp-content/uploads/sites/412/2025/09/11175053/The-Myths-of-the-Last-Supper.pdf> (visited on 08/21/2026).

³³Ibid.

³⁴Ibid.

³⁵WBUR, *The Last Supper: How a 1993 Pentagon Dinner Reshaped the Defense Industry*, op. cit.; Mintz, “How a Dinner Led to a Feeding Frenzy,” op. cit.

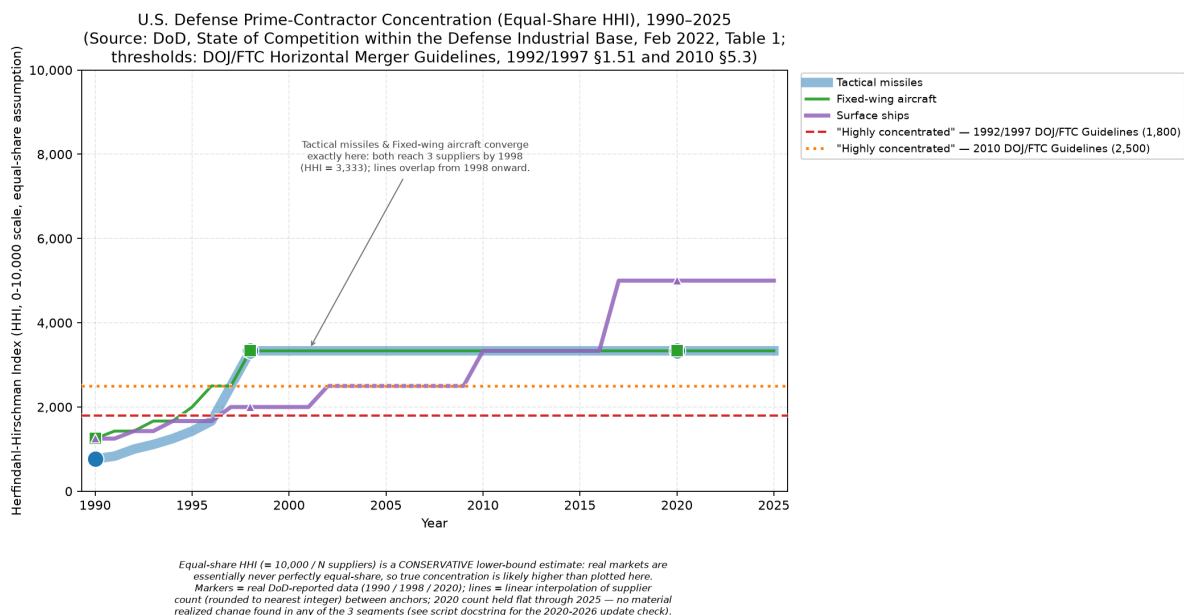


Figure 2: Three core weapons categories crossed the DOJ/FTC “highly concentrated” threshold by the early 2000s and have not recrossed back since.

Source: Prime-contractor counts from U.S. Department of Defense, *State of Competition within the Defense Industrial Base* (Feb. 2022), Table 1; index computed under a conservative equal-share assumption (10,000/ N), a documented floor on true concentration, not a point estimate.

a broken market.³⁶ RTX derives only 59 percent of its revenue from defense work and Boeing 44 percent; the top five primes together capture less than 20 percent of the total defense budget, a smaller share than Walmart, Amazon, and CVS hold of retail.³⁷ More than 60,000 companies and 25,000 direct-award recipients populate the sub-tier base; small businesses made up 73 percent of DoD’s contracting firms in FY2021.³⁸ He models DoD as a monopsony “Visible Hand” managing a portfolio, not a textbook market.³⁹

Both critics are answered the same way: neither changes what DoD itself publishes.⁴⁰ The question was never how many companies call themselves defense contractors, or how diversified their books look. It is who owns each weapons category, and DoD’s own count, not this paper’s, shows one to four suppliers in every one that matters. The chokepoints those numbers describe, in missiles, satellites, surface ships, and tracked vehicles, are where the evidence turns next.

³⁶ John D. Driessnack. *Defense Market(s): A Relook at the Explanatory Power of Several Economic Schools of Thought while Viewing the Defense Department as a Monopsony Firm*. SYM-AM-25-362. Naval Postgraduate School, Acquisition Research Program, May 5, 2025. URL: <https://dair.nps.edu/bitstream/123456789/5414/1/SYM-AM-25-362.pdf> (visited on 08/21/2026).

³⁷Ibid.

³⁸Ibid.

³⁹Ibid.

⁴⁰U.S. Department of Defense, *State of Competition within the Defense Industrial Base*, op. cit.

3 The Bill Comes Due

The 2025–26 reforms arrived with real money behind them. Fiscal year 2026 national-defense discretionary spending reached \$1.05 trillion, the first nominal trillion-dollar defense budget in U.S. history.⁴¹ The Congressional Budget Office found that base-budget acquisition funding, the durable dollars a supplier actually builds a factory around, fell 9.0 percent even as one-time reconciliation money pushed the acquisition topline up 18 percent.⁴² Money is moving. The industrial base receiving it has not changed shape. Four features of that unchanged base explain why the bill keeps coming due.

The Chokepoint Map

Seven weapons segments show where competition actually disappeared. Six of the seven named in DoD’s own 2022 State of Competition report and subsequent GAO reviews carry one or two qualified U.S. suppliers apiece (Table 2).⁴³ Solid rocket motors, the propulsion behind every U.S. tactical and strategic missile, fell from roughly six manufacturers in 1995 to two today: L3Harris, which absorbed Aerojet Rocketdyne on July 28, 2023, and Northrop Grumman. GAO found the sector’s own supplier base shrank from about 5,000 firms to 1,000 over twenty years.⁴⁴

The pattern repeats elsewhere in the table. Huntington Ingalls is the sole U.S. builder of nuclear-powered aircraft carriers, whose lead Ford-class ship grew 23 percent over its original cost estimate, and BAE Systems’ M777 towed-howitzer line sits idle, the company saying it would need at least 150 new orders to restart it.⁴⁵ Regulators have recognized the scarcity before. In January 2022 the FTC sued to block Lockheed Martin’s \$4.4 billion acquisition of Aerojet Rocketdyne, then the country’s last independent solid-rocket-motor maker, arguing a major missile integrator could foreclose rival primes from the same propulsion supply; Lockheed abandoned the deal nineteen days later, the first FTC-litigated defense-merger challenge in decades.⁴⁶

Sole-source status is a price, not an abstraction. Two Pentagon inspector-general audits

⁴¹Arms Control Today. *U.S. Defense Spending Rises by More Than 17 Percent*. Mar. 2026. URL: <https://www.armscontrol.org/act/2026-03/news/us-defense-spending-rises-more-17-percent> (visited on 08/21/2026).

⁴²Congressional Budget Office. *DoD’s 2026 Budget Request and Plan for Funding Provided by the 2025 Reconciliation Act*. Congressional Budget Office, Apr. 2026. URL: <https://www.cbo.gov/system/files/2026-04/62301-DoD-2026-Budget-Request.pdf> (visited on 08/21/2026).

⁴³U.S. Department of Defense, *State of Competition within the Defense Industrial Base*, op. cit.

⁴⁴U.S. Government Accountability Office. *Solid Rocket Motors: DOD and Industry Are Addressing Challenges to Minimize Supply Concerns*. GAO-18-45. U.S. Government Accountability Office, Oct. 26, 2017. URL: <https://www.gao.gov/assets/gao-18-45.pdf> (visited on 08/21/2026).

⁴⁵U.S. Government Accountability Office. *Ford-Class Aircraft Carrier: Follow-On Ships Need More Frequent and Accurate Cost Estimates to Avoid Pitfalls of Lead Ship*. GAO-17-575. U.S. Government Accountability Office, June 13, 2017. URL: <https://www.gao.gov/assets/gao-17-575.pdf> (visited on 08/21/2026); Jones et al., *Empty Bins in a Wartime Environment: The Challenge to the U.S. Defense Industrial Base*, op. cit.

⁴⁶Federal Trade Commission. *FTC Sues to Block Lockheed Martin Corporation’s \$4.4 Billion Vertical Acquisition of Aerojet Rocketdyne*. Jan. 25, 2022. URL: <https://www.ftc.gov/news-events/news/press-releases/2022/01/ftc-sues-block-lockheed-martin-corporations-44-billion-vertical-acquisition-aerojet-rocketdyne> (visited on 08/21/2026).

Segment	Supplier(s)	<i>N</i>	Status
Solid rocket motors	L3Harris; Northrop Grumman	2	Duopoly, down from 6 (1995)
Nuclear submarines	GD Electric Boat; HII Newport News	2	Capacity-constrained
ICBM (Sentinel)	Northrop Grumman	1	Sole-source
Patriot PAC-3 MSE interceptor	Lockheed Martin	1	Sole-source integrator
Towed 155mm howitzer (M777)	BAE Systems	1	Sole-source; line idle
Aircraft carriers	HII Newport News	1	Sole-source
155mm shell production	GD-OTS; American Ordnance; IMT; Nammo	4	Diversified since 2022; one line stalled

Table 2: Six of seven mapped chokepoint segments carry one or two qualified U.S. suppliers each; only munitions assembly shows genuine multi-sourcing, and even that remains fragile.

Source: U.S. Department of Defense, *State of Competition within the Defense Industrial Base* (Feb. 2022); GAO-18-45, GAO-24-107732, and GAO-17-575; U.S. Air Force and Army program reporting; CSIS, *Empty Bins in a Wartime Environment* (Jan. 2023). Full citations in text.

of TransDigm, a firm that specializes in buying companies that hold sole-source spare-parts contracts, found the pattern directly. The first, in 2019, identified \$16.1 million in “excess profit” on 46 of 47 sampled parts, with margins as high as 4,451 percent.⁴⁷ A follow-on audit released in December 2021 found at least \$20.8 million more in excess profit on 105 of 106 parts.⁴⁸ Both audits found the pricing legal: a monopsony buyer with no alternative supplier has no leverage to negotiate a lower one.

The Sentinel intercontinental ballistic missile (ICBM) shows the same dynamic at prime-contractor scale. After Boeing withdrew as a competing bidder in 2020, the Air Force awarded Northrop Grumman a sole-source contract, and the program’s total cost estimate has since grown to \$140.9 billion, an 81 percent increase over the 2020 baseline, a breach severe enough to trigger a critical Nunn-McCurdy review with no competing bid anywhere in the process to check the number.⁴⁹ Sole-source pricing power is not confined to spare parts; it reaches the prime contract itself, where no competing bid exists to check the government’s own number.

⁴⁷U.S. Department of Defense, Office of Inspector General. *Review of Parts Purchased From TransDigm Group, Inc.* DODIG-2019-060. U.S. Department of Defense, Office of Inspector General, Feb. 25, 2019. URL: <https://media.defense.gov/2019/Feb/27/2002093922/-1/-1/1/DODIG-2019-060.PDF> (visited on 08/21/2026).

⁴⁸U.S. Department of Defense, Office of Inspector General. *Audit of the Business Model for TransDigm Group Inc. and Its Impact on Department of Defense Spare Parts Pricing.* DODIG-2022-043. U.S. Department of Defense, Office of Inspector General, Dec. 13, 2021. URL: <https://media.defense.gov/2021/Dec/27/2002914678/-1/-1/1/DODIG-2022-043%20508.PDF> (visited on 08/21/2026).

⁴⁹Air Force News Service. *Department of Defense Announces Results of Sentinel Nunn-McCurdy Review.* U.S. Department of the Air Force. July 8, 2024. URL: <https://www.af.mil/News/Article-Display/Article/3830205/> (visited on 08/21/2026).

Surge Failure

The 155mm artillery-shell ramp is the clearest test of whether a concentrated base can surge. The Army began 2022 producing about 14,000 rounds a month; production has since risen to 40,000, but every one of six Army and Pentagon targets announced between December 2022 and July 2025, each aiming for 60,000 to 100,000 rounds a month, has slipped, most recently from October 2025 to mid-2026 (Figure 3).⁵⁰ A propellant shortage cut September 2024 output to just 18,000 complete rounds, well under plan.⁵¹ The Pentagon's \$5 billion investment in new capacity shows the same pattern at larger scale: General Dynamics' \$469 million automated shell-casing plant in Mesquite, Texas, had produced no qualifying subcomponents at all as of March 2026, more than two years after the investment.⁵² As late as April 2025 the Army was still working to eliminate the last single-source suppliers of 155mm components, an admission that three years into a wartime surge, sole-sourcing had not been solved.⁵³

Patriot tells the same story from the demand side. Lockheed Martin, the sole integrator of the Patriot Advanced Capability-3 Missile Segment Enhancement (PAC-3 MSE) interceptor, raised its contracted rate from 550 to 650 missiles a year in late October 2024 and delivered a production-record 620 interceptors in 2025.⁵⁴ By July 2025, U.S. Patriot stocks had nonetheless fallen to just 25 percent of the Pentagon's own minimum requirement, driven by transfers to Ukraine.⁵⁵ A January 2026 framework agreement targets 2,000 interceptors a year by 2030, more than triple current output, a build-out that will take years even with the money already committed.⁵⁶ Smaller systems show the same lag on a longer fuse: Javelin transfers to Ukraine through August 2022 alone equaled seven years of production at the sole-source rocket motor's 2022 rate, and fielding a next-generation Stinger replacement, rather than restarting a semi-idle line, would take five to seven years, too slow, in CSIS's own judgment, to matter in a near-term fight.⁵⁷ The pattern holds at every scale, from a \$5 billion capacity-expansion program to a shoulder-fired missile line: a concentrated base can spend money faster than it can grow

⁵⁰Defense News. *Army Races to Widen the Bottlenecks of Artillery Shell Production*. Oct. 14, 2024. URL: <https://www.defensenews.com/land/2024/10/14/army-races-to-widen-the-bottlenecks-of-artillery-shell-production/> (visited on 08/21/2026); Breaking Defense. *Army 'Hitting Stride' with 155mm Production, but Key General Worries over What's Needed Next*. Oct. 17, 2025. URL: <https://breakingdefense.com/2025/10/army-hitting-stride-with-155mm-production-but-key-general-worries-over-whats-needed-next/> (visited on 08/21/2026).

⁵¹Defense News, *Army Races to Widen the Bottlenecks of Artillery Shell Production*, op. cit.

⁵²ProPublica. *General Dynamics Artillery Factory Failed*. July 14, 2026. URL: <https://www.propublica.org/article/general-dynamics-artillery-factory-failed> (visited on 08/21/2026).

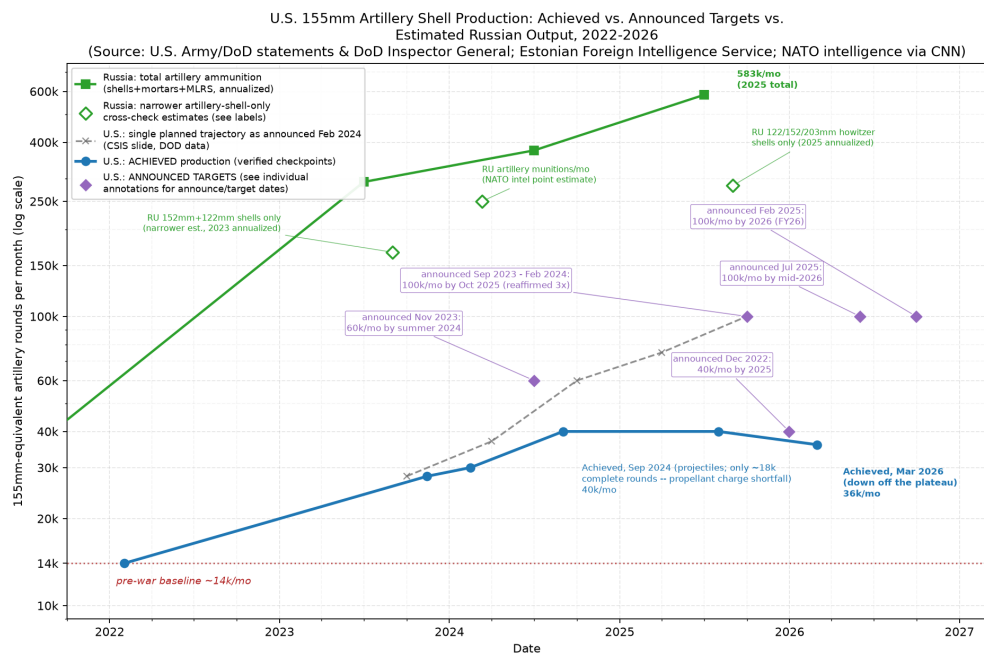
⁵³U.S. Army. *Army Seeks to Expand and Accelerate 155 mm Production*. Apr. 4, 2025. URL: https://www.army.mil/article/283210/army_seeks_to_expand_and_accelerate_155_mm_production (visited on 08/21/2026).

⁵⁴U.S. Army. *US Army Awards Patriot Advanced Capability-3 Contract Worth Up to \$368.6 Million*. Nov. 27, 2024. URL: https://www.army.mil/article/281669/us_army_awards_patriot_advanced_capability_3_contract_worth_up_to_368_6_million (visited on 08/21/2026).

⁵⁵Foreign Policy Research Institute. *Scaling Patriot Production: The Industrial Base Crisis Explained*. June 23, 2026. URL: <https://www.fpri.org/article/2026/05/scaling-patriot-production-the-industrial-base-crisis-explained/> (visited on 08/21/2026).

⁵⁶Ibid.

⁵⁷Jones et al., *Empty Bins in a Wartime Environment: The Challenge to the U.S. Defense Industrial Base*, op. cit.



Note: Russian figures are annual totals / 12, plotted at mid-year; "total" includes mortar rounds and MLRS rockets, not shells alone. The "122/152/203mm howitzer" and "152mm+122mm" markers are the closer artillery-shell-only proxies for 155mm-equivalent comparison. U.S. Sep-2024 figure counts complete rounds far lower than projectile-body count (propellant-charge bottleneck).

Figure 3: Three years and \$5 billion in new capacity nearly tripled monthly shell output, from 14,000 rounds in early 2022 to 40,000 by June 2025, yet every 60,000–100,000-round target the Army and Pentagon have set since December 2022 has slipped, most recently to mid-2026. Source: Defense News (2024), Breaking Defense (2025), and ProPublica (2026) production reporting, cited in text; chart generated by `analysis/surge_ramp.py`.

suppliers. Surge, on this base, arrives late.

Brittle Precision, Cheap Attackers

Two and a half years of Russian jamming exposed a second failure mode: the same consolidated base that struggles to surge volume also cannot cheaply harden precision. The Excalibur artillery shell, guided by the Global Positioning System (GPS), saw its hit rate collapse from 55 percent in January 2023 to 6 percent by August 2023, a decline compiled from nearly 3,000 rounds fired from U.S.-supplied M777 howitzers and documented in confidential Ukrainian military reports shared with Washington; Ukraine stopped using the round.⁵⁸ The same jamming pushed Guided Multiple Launch Rocket System (GMLRS) rockets fired from High Mobility Artillery Rocket System (HIMARS) launchers to miss by 15 meters or more, eroding a system celebrated a year earlier for precision strikes on Russian depots and command posts.⁵⁹ The R-330Zh Zhitel, a vehicle-mounted jammer that blocks GPS signals at 30 kilometers, is the system most directly

⁵⁸Kyiv Post. *Analysis: Off Target – Are Kyiv’s GPS-Aided Weapons Susceptible to Kremlin EW Jamming?* May 29, 2024. URL: <https://www.kyivpost.com/analysis/33448> (visited on 08/21/2026).

⁵⁹The Washington Post. *Russian Jamming of U.S. Weapons in Ukraine Forces Pentagon to Adjust*. May 24, 2024. URL: <https://www.washingtonpost.com/world/2024/05/24/russia-jamming-us-weapons-ukraine/> (visited on 08/21/2026).

implicated in both failures.⁶⁰ The Air Force’s fix, a “home-on-GPS-jam” seeker for the Joint Direct Attack Munition-Extended Range (JDAM-ER) glide bomb contracted in May 2024, took seventeen months to reach its October 2025 fielding target, an unusually fast turnaround by U.S. acquisition standards.⁶¹ Excalibur, GMLRS, and JDAM-ER are exactly the low-volume, high-cost precision munitions a five-prime base is built to produce. Shared GPS dependency is a common-mode failure, and a seventeen-month fix is a poor answer to a jammer built for a few hundred dollars.

Cost-exchange asymmetry compounds the problem. A Shahed-136/Geran-2 one-way attack drone costs Russia an estimated \$20,000 to \$70,000 to build; the interceptors the United States and its partners fire back run from \$1.8 million for an Evolved Sea Sparrow Missile to \$5.3 million for a Patriot PAC-3 MSE, ratios of roughly 35:1 to 265:1 in the attacker’s favor (Figure 4).⁶² At the high end, scaled to everyday prices, that ratio is a \$5,300 shot fired at a \$20 target. Over roughly fifteen months of Red Sea operations against Houthi drones and missiles, the U.S. Navy fired about 220 interceptors at a cost near \$1 billion to defeat 380 targets.⁶³

The ratio holds only until a raid outgrows a defending ship’s interceptor magazine; past that point unintercepted drones get through regardless of cost, which is why the cheapest attacker win in this record came without a single missed intercept. On January 28, 2024, a one-way drone approached Tower 22 in Jordan at the same moment a U.S. surveillance drone was returning to base. The resulting identification confusion delayed a response that never came, killing three U.S. soldiers and wounding 47.⁶⁴ The cost-exchange framing is not the whole picture: CSIS’s Wes Rumbaugh has argued that defensive intercept is an inherently harder engineering problem than one-way attack flight, and that the relevant comparison is the value of the defended asset, not sticker price alone.⁶⁵ Neither correction changes the conclusion that a base built for exquisite, expensive interceptors has no cheap answer to mass.

⁶⁰Thomas Withington. *Blunting Excalibur’s Edge*. European Security & Defence. July 31, 2024. URL: <https://euro-sd.com/2024/07/articles/39533/blunting-excaliburs-edge/> (visited on 08/21/2026).

⁶¹The Defense Post. *USAF Buys Add-On JDAM-ER Seeker to Target Russian GPS Jammers*. May 7, 2024. URL: <https://thedefensepost.com/2024/05/07/usaf-jdam-seeker-russian/> (visited on 08/21/2026).

⁶²Defense Express. *How Much Russia Pays for Shahed Now, Down From \$300,000 per Drone in 2022*. Aug. 24, 2025. URL: https://en.defence-ua.com/industries/how_much_russia_pays_for_shahed_now_down_from_300000_per_drone_in_2022-15585.html (visited on 08/21/2026); Wes Rumbaugh. *Cost and Value in Air and Missile Defense Intercepts*. Center for Strategic and International Studies, Feb. 13, 2024. URL: <https://www.csis.org/analysis/cost-and-value-air-and-missile-defense-intercepts> (visited on 08/21/2026); The War Zone. *‘Cheap’ Patriot Interceptor Costing Under \$1 Million Now Being Sought by Army*. May 18, 2026. URL: <https://www.twz.com/land/cheap-patriot-interceptor-costing-under-1-million-now-being-sought-by-army> (visited on 08/21/2026).

⁶³Stars and Stripes. *Houthis’ Red Sea Attacks Have Cost the U.S. Navy Nearly \$1 Billion in Munitions*. Jan. 16, 2025. URL: <https://www.stripes.com/branches/navy/2025-01-16/houthis-navy-red-sea-missiles-drones-16500246.html> (visited on 08/21/2026).

⁶⁴CNN. *Tower 22: Drone That Killed US Soldiers in Jordan Followed American Drone onto Base, Causing Confusion*. Jan. 30, 2024. URL: <https://www.cnn.com/2024/01/29/politics/us-soldiers-deaths-jordan/index.html> (visited on 08/21/2026).

⁶⁵Rumbaugh, *Cost and Value in Air and Missile Defense Intercepts*, op. cit.

Interceptor-vs-Drone Cost Ratio Declines Once Salvo Size Exceeds Magazine Depth
(Source: Stars and Stripes Jan 2025; TWZ Dec 2020 & May 2026; CSIS Feb 2024; CRS RL32109 -- see script header)

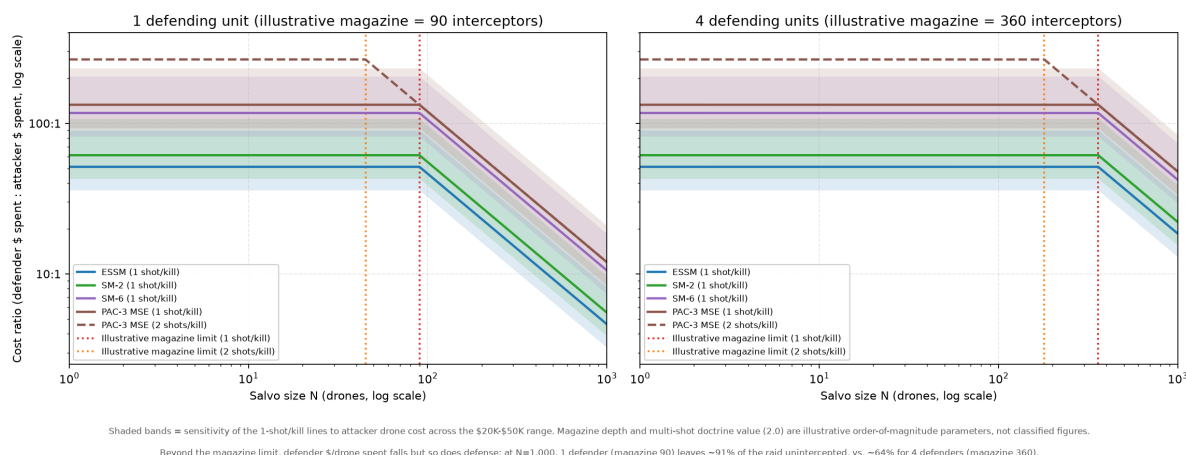


Figure 4: Interceptors cost 35 to 265 times as much as the Shahed-class drones they defeat at a fixed exchange rate, a ratio that matters only until a raid outgrows the defender’s magazine depth.
Source: unit-cost reporting from Defense Express (2025), CSIS (2024), The War Zone (2026), and Stars and Stripes (2025), cited in text; chart generated by `analysis/cost_exchange.py`.

Against the Pacing Threat

The industrial base sized against two decades of counterinsurgency has not been resized against China. In 24 iterations of a CSIS wargame modeling a Chinese invasion of Taiwan, the United States expended its entire global inventory of Long Range Anti-Ship Missiles (LRASMs), about 450 missiles, within the first week of the conflict in every single iteration, against a production rate of just 88 missiles planned for fiscal year 2023.⁶⁶ The bottleneck was a portfolio choice, not a technology gap. LRASM and the general-purpose Joint Air-to-Surface Standoff Missile (JASSM) share 70 percent of their parts and the same production line, yet the FY2023 plan called for 581 JASSMs against just 88 LRASMs, the inverse of wartime demand and, in CSIS’s own words, “backward.”⁶⁷ Three years later, the Heritage Foundation’s January 2026 TIDALWAVE assessment found the gap essentially unchanged: fewer than 450 LRASMs in the global inventory against a congressionally derived requirement of 1,000 to 1,200, with annual production still near 115 missiles.⁶⁸

By prime-contractor count, China’s defense industry is more concentrated than America’s: roughly seven state conglomerates against the United States’ five.⁶⁹ Its edge instead comes from capacity and civil-commercial depth: CSIS puts China’s shipbuilding capacity at roughly

⁶⁶Cancian, Cancian, and Heginbotham, *The First Battle of the Next War: Wargaming a Chinese Invasion of Taiwan*, op. cit.

⁶⁷Ibid.

⁶⁸The Heritage Foundation, *TIDALWAVE, Chapter 6: Assessing the U.S. Indo-Pacific Munitions System*, op. cit.

⁶⁹Cortez A. Weinbaum et al. *Assessing Systemic Strengths and Vulnerabilities of China’s Defense Industrial Base*. RB-A930-1. RAND Corporation, Feb. 2022. URL: https://www.rand.org/content/dam/rand/pubs/research_briefs/RBA900/RBA930-1/RAND_RBA930-1.pdf (visited on 08/21/2026).

230 times the U.S. level and its rate of acquiring high-end weapons and munitions at five to six times the U.S. pace.⁷⁰ The same civil-commercial depth shows up in consumer hardware, where Chinese-made systems hold roughly 70 percent of the global drone market and more than 90 percent of the U.S. commercial drone market, led overwhelmingly by one firm, DJI.⁷¹ Military-Civil Fusion policy, not a larger competitor roster, is the mechanism: the same shipyards that build container ships also build destroyers.⁷² A pacing threat measured in weeks of wartime consumption cannot be met by a base whose only lever, for three decades, has been fewer and larger primes.

4 Competition Works — Under Conditions

Deconsolidation is not a bet on competition in the abstract. The record spans an American engine program, a decade of Cold War missile contracts, a wartime drone industry, and a Soviet design-bureau system the Kremlin itself later dismantled. Read together, the record is conditional: dual-sourcing paid when volume and learning-curve economics allowed it and failed when they did not, and distributed production beat consolidated production only where something removed the normal peacetime barriers to entry. That conditionality is the discipline the remedy below has to keep: run the math program by program, not once for the whole base.

The Great Engine War

The Great Engine War is dual-sourcing’s best-documented success, and it worked for reasons a program office can check in advance, not because competition is inherently cheaper. Pratt & Whitney’s F100 engine, sole-sourced since 1974 for the F-15 and F-16, developed in-flight compressor stalls (“stall-stagnation”) and short-life, high-maintenance problems severe enough that the Air Force went looking for an alternative. General Electric self-funded a demonstrator in 1975, and Congress funded a derivative-engine program starting in fiscal year (FY) 1979 over Pratt & Whitney’s objection.⁷³ In February 1984 the Air Force made its first competitive award: 120 F110 engines, 75 percent of the buy, to GE for the F-16, and 40 upgraded F100-PW-220 engines, 25 percent, to Pratt & Whitney for the F-15, re-competed annually through FY1990.⁷⁴ Both engines were derivatives of existing designs, not clean-sheet developments, which RAND

⁷⁰Seth G. Jones and Alexander Palmer. *China Outpacing U.S. Defense Industrial Base*. Center for Strategic and International Studies, Mar. 6, 2024. URL: <https://www.csis.org/analysis/china-outpacing-us-defense-industrial-base> (visited on 08/21/2026).

⁷¹U.S.-China Economic and Security Review Commission. *Made in China 2025: Evaluating China’s Performance*. U.S.-China Economic and Security Review Commission, Nov. 14, 2025. URL: https://www.uscc.gov/sites/default/files/2025-11/Made_in_China_2025--Evaluating_Chinas_Performance.pdf (visited on 08/21/2026).

⁷²Weinbaum et al., *Assessing Systemic Strengths and Vulnerabilities of China’s Defense Industrial Base*, op. cit.

⁷³Frank Camm and Jr. Glennan Thomas K. *The Development of the F100-PW-220 and F110-GE-100 Engines: A Case Study of Risk Assessment and Risk Management*. Note N-3618-AF. RAND Corporation, 1993. URL: <https://www.rand.org/content/dam/rand/pubs/notes/2007/N3618.pdf> (visited on 08/21/2026).

⁷⁴U.S. General Accounting Office. *Review of the Alternate Fighter Engine Competition*. GAO/NSIAD-84-104. U.S. General Accounting Office, Apr. 27, 1984. URL: <https://www.gao.gov/products/nsiad-84-104> (visited on 08/21/2026).

identifies as the reason entry cost stayed low: joint government research and development for both programs totaled about \$610 million over seven years, against an estimated \$1.5 billion for a single new “centerline” engine.⁷⁵

The Air Force and the Government Accountability Office (GAO) projected in April 1984 that the competition could save up to \$3 billion over 20 years against continued sole-source procurement.⁷⁶ A quarter-century later, GAO’s own retrospective found the program had delivered nearly 30 percent cumulative acquisition-cost savings, roughly 16 percent operations-and-support savings, and 21 percent savings in total life-cycle cost.⁷⁷ RAND’s own performance data show reliability moved with the price: cost per flight hour fell 41 to 63 percent, unscheduled engine removals fell 94 to 96 percent per thousand flight hours, and the interval between depot overhauls roughly doubled.⁷⁸ The decisive lever was contractual as much as competitive: a 1983 appropriations provision required a minimum 3,000-cycle warranty on any new fighter engine, and as Pratt & Whitney’s award share fell, its warranty price rose as much as threefold while GE’s fell, an asymmetry GAO credited as the reason the 1985 award split rather than went to a single winner.⁷⁹

The Air Force never repeated the experiment. For the F119 engine that followed on the F-22, it judged a clean-sheet second engine too expensive to duplicate and stayed sole-source, and RAND’s own retrospective warned that carrying two engines through full development “essentially doubled the cost of creating the technological capability” the derivatives offered.⁸⁰ The Great Engine War worked because of specific, quantifiable conditions that will not automatically recur elsewhere in the portfolio: derivative engines cheap enough to qualify a second line, a warranty lever strong enough to move price with reliability, and enough remaining production running to amortize the fixed cost of standing up a competitor. The wider record is less forgiving.

The Honest Record

Outside the Great Engine War, the dual-sourcing record turns decisively mixed, and a paper resting a remedy on this evidence has to say so plainly. The concession narrows the remedy’s reach without erasing the case for it. An unpublished 1987 Office of the Secretary of Defense (OSD) analysis tested three cost-estimation methods against actual contract prices for four dual-sourced missile lines, the AIM-7F, AIM-7M, AIM-9L, and AIM-9M Sparrow and Sidewinder variants. Its finding was blunt: dual-sourcing cost 11 to 42 percent more than an estimated

⁷⁵Camm and Glennan, *The Development of the F100-PW-220 and F110-GE-100 Engines: A Case Study of Risk Assessment and Risk Management*, op. cit.

⁷⁶U.S. General Accounting Office, *Review of the Alternate Fighter Engine Competition*, op. cit.

⁷⁷Jeremiah Gertler. *F-35 Alternate Engine Program: Background and Issues for Congress*. R41131. Congressional Research Service, Jan. 10, 2012. URL: https://www.congress.gov/crs_external_products/R/PDF/R41131/R41131.15.pdf (visited on 08/21/2026).

⁷⁸Camm and Glennan, *The Development of the F100-PW-220 and F110-GE-100 Engines: A Case Study of Risk Assessment and Risk Management*, op. cit.

⁷⁹U.S. General Accounting Office, *Review of the Alternate Fighter Engine Competition*, op. cit.

⁸⁰Camm and Glennan, *The Development of the F100-PW-220 and F110-GE-100 Engines: A Case Study of Risk Assessment and Risk Management*, op. cit.

sole-source alternative in 11 of 12 method-program combinations, and the sole exception showed a marginal 1.1 percent saving.⁸¹ RAND’s independent five-method reanalysis of those same four programs reached the same mixed, largely negative conclusion.⁸²

The best correction for this scatter is the only peer-reviewed econometric treatment in the literature. Thomas Lyon’s 2006 study in the *Journal of Industrial Economics* panel-analyzed 14 tactical-missile programs and 175 system-years from 1975 to 1995. A naive before-and-after comparison, the method most 1980s program offices actually used, found a statistically insignificant 4.6 percent saving from dual sourcing; correcting for the fact that the Pentagon chose which programs got a second source, using a Heckman selection correction, Lyon found the true average saving was 20.4 percent, and significant. Selection bias alone had masked the real effect by a factor of four.⁸³

RAND’s Birkler school converted this scatter into a decision rule rather than a debate. Its Required Cost Reduction (RCR) formula computes the price cut a second source must sustain for the government to break even; a result above 100 percent means savings are mathematically impossible under the stated assumptions.⁸⁴ Applied to RAND’s own historical survey, missile and ship programs cleared a median of only about 17 percent realized savings and none cleared 40 percent; electronics programs did better, with a median near 30 percent and 4 of 10 programs clearing 40 percent. No missile or ship program with a learning-curve slope flatter than 85 to 90 percent, and no electronics program flatter than 80 percent, has ever proved worth dual-sourcing.⁸⁵ The threshold, not the average, is the rule.

Figure 5 puts a dollar shape on that conditionality, anchored to the same GAO figures behind the Great Engine War and a Congressional Budget Office learning-curve baseline. Modeled over a ten-year, 27,000-unit missile program under Wright’s-law learning curves, dual-sourcing’s price advantage is likely to fall near a point estimate of 21 percent of unit cost by year 10, matching the Great Engine War’s own life-cycle figure, but the honestly reportable range runs from 10 to 30 percent depending on where the true learning-curve slope and competition discount actually fall.⁸⁶ That range, not the point estimate, is the honest answer to whether dual-sourcing pays: it depends on volume, on how flat the incumbent’s own learning curve already is, and on how much production remains to amortize a second qualified line.

That conditionality is what makes this evidence usable, not the average across every program

⁸¹Office of the Secretary of Defense, Program Analysis and Evaluation. *Dual Sourcing in Defense Missile Procurement*. Unpublished draft, the “Missile Paper”. U.S. Department of Defense, Apr. 8, 1987. URL: https://economics.osd.mil/dual_sourcing.pdf (visited on 08/21/2026).

⁸²John Birkler, Edmund Dews, and Joseph P. Large. *Issues Associated with Second-Source Procurement Decisions*. R-3996-RC. RAND Corporation, 1990. URL: <https://www.rand.org/content/dam/rand/pubs/reports/2007/R3996.pdf> (visited on 08/21/2026).

⁸³Thomas P. Lyon. “Does Dual Sourcing Lower Procurement Costs?” In: *Journal of Industrial Economics* 54.2 (2006), pp. 223–252. URL: <https://webuser.bus.umich.edu/tplyon/PDF/Published%20Papers/Lyon%20JIE%20Dual%20Sourcing.PDF> (visited on 08/21/2026).

⁸⁴Mark V. Arena and John Birkler. *Determining When Competition Is a Reasonable Strategy for the Production Phase of Defense Acquisition*. Occasional Paper OP-263. RAND Corporation, 2009. URL: https://www.rand.org/content/dam/rand/pubs/occasional_papers/2010/RAND_OP263.pdf (visited on 08/21/2026).

⁸⁵*Ibid.*

⁸⁶U.S. General Accounting Office, *Review of the Alternate Fighter Engine Competition*, op. cit.; Gertler, *F-35 Alternate Engine Program: Background and Issues for Congress*, op. cit.

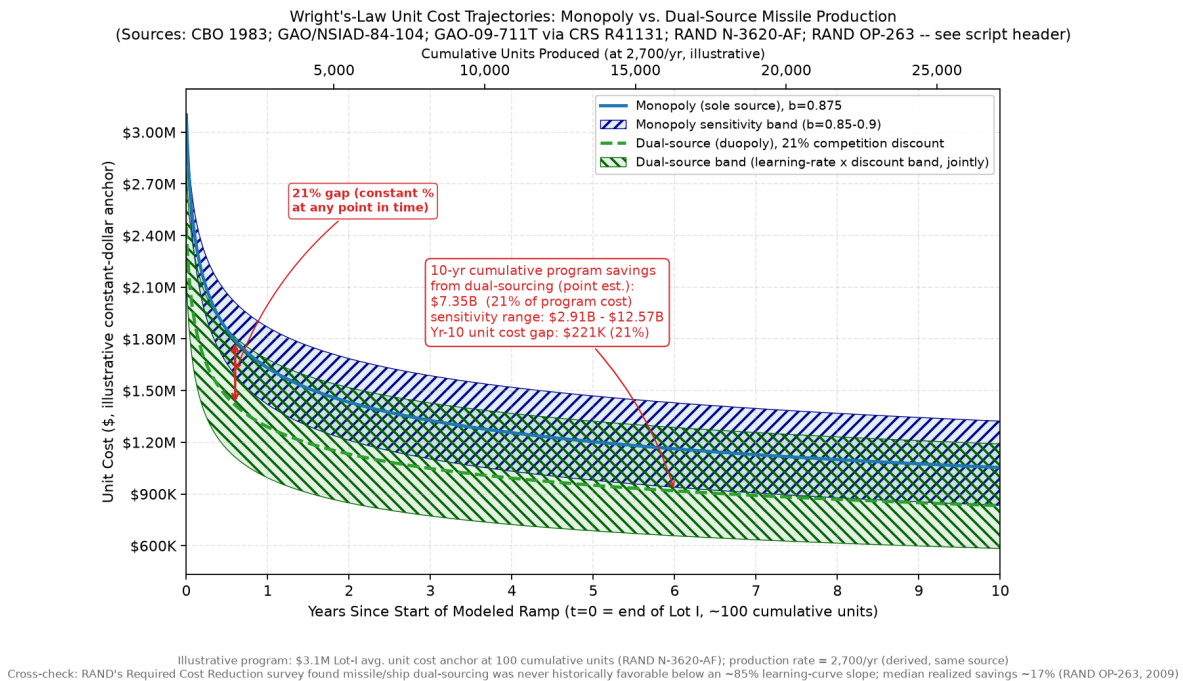


Figure 5: Dual-sourcing's modeled price advantage over a monopoly baseline reaches a point estimate of 21 percent of unit cost by year 10 of a 27,000-unit missile program, with an honest sensitivity band of 10–30 percent across the underlying learning-curve and competition-discount ranges.

Source: Author's calculations (Wright's-law model), anchored to GAO/NSIAD-84-104 (1984) and GAO-09-711T (2009) via CRS R41131, with learning-curve parameters from CBO (1983).

RAND ever studied. RAND's breakeven math tells a policymaker exactly where the savings are before a program office signs a second contract, and a remedy that runs that math segment by segment, rather than mandating a second source everywhere, is the version this record actually supports. The remedy that follows runs on this math.

Ukraine's Existence Proof

Ukraine ran the same experiment in reverse, and proved the same point under fire, without a model to follow. Before Russia's 2022 invasion, drone production sat inside the state monopoly Ukroboronprom and independent manufacturers numbered roughly seven to ten.⁸⁷ Ukrainian government figures counted an estimated 500 dedicated drone manufacturers by 2025, and by August 2026 the government's own Brave1 defense-technology cluster listed 2,500 member companies across drone, electronic-warfare, ground-robotics, and munitions categories.⁸⁸

⁸⁷Kateryna Bondar. *How Ukraine Rebuilt Its Military Acquisition System Around Commercial Technology*. Center for Strategic and International Studies, Jan. 13, 2025. URL: https://csis-website-prod.s3.amazonaws.com/s3fs-public/2025-01/250113_Bondar_Ukraine_Acquisition.pdf (visited on 08/21/2026); Army Recognition. *Ukraine Emerges as World Leader in Drone Technology Driven by Battle-Proven Innovation*. Nov. 13, 2025. URL: <https://www.armyrecognition.com/news/aerospace-news/2025/ukraine-emerges-as-world-leader-in-drone-technology-driven-by-battle-proven-innovation> (visited on 08/21/2026).

⁸⁸Brave1. *Brave1 Defense Tech Cluster*. URL: <https://brave1.gov.ua/en> (visited on 08/21/2026).

Ukraine’s finding is speed, not headcount: how fast a battlefield failure becomes a new prototype. Under the Soviet-legacy system Ukraine inherited, concept-to-prototype for a new system took three to twenty years, and adoption into service took another one to two years. Under the wartime, private-sector-led system that replaced it, unmanned-systems and electronic-warfare concepts move from idea to fielded prototype in a few weeks to six months, with testing and codification compressed to two weeks.⁸⁹ The American comparison is not flattering: GAO’s most recent weapons-assessment report found the average time for a Major Defense Acquisition Program (MDAP) to reach even initial capability rose 18 months in a single year, to almost 12 years from program start, and that even the Pentagon’s fast-track “Middle Tier of Acquisition” pathway, built for a two-to-five-year timeline, routinely exceeds it.⁹⁰

This is an existence proof, not a template for direct transplant. Ukraine’s ecosystem runs on a wartime demand signal no peacetime American program can replicate, and on a regulatory floor that sits close to zero. What it proves is narrower and still load-bearing: distributed, competitively selected production converts industrial capacity into fielded output far faster than a handful of consolidated primes can, whenever the state gets out of the way enough to let it. Structure did that, not process.

The Precedent and the Counterexample

Even a centrally planned command economy sustained rival weapons-design centers, which is the strongest precedent for reintroducing competition into a base already trending toward monopoly. The Soviet General Staff’s 1969 requirement for an “Advanced Frontline Fighter” split in 1971 into a heavy and a light variant: Sukhoi’s design became the Su-27, entering service in 1985, and Mikoyan’s became the MiG-29, reaching combat units in 1983, and both aircraft were built and fielded rather than one being cancelled once a winner was picked.⁹¹ From 1954 to 1964, three rival chief designers, Sergei Korolev, Vladimir Chelomei, and Mikhail Yangel, ran independent missile and space bureaus; Yangel’s was split off from Korolev’s in 1954 for an explicit dispersal-for-survivability rationale, so a single strike on Moscow could not destroy the country’s entire missile-design capacity, and a 3 August 1964 Central Committee decree split the lunar program three ways among the same rivals.⁹²

The Soviets conceded the cost of this rivalry themselves, which bounds how far the precedent can be pushed. A declassified 1988 Central Intelligence Agency (CIA) assessment found true Western-style prototype fly-offs were “vestiges ... occasionally still apparent” and mostly

⁸⁹Bondar, *How Ukraine Rebuilt Its Military Acquisition System Around Commercial Technology*, op. cit.

⁹⁰U.S. Government Accountability Office. *Weapon Systems Annual Assessment*. GAO-25-107569. U.S. Government Accountability Office, June 11, 2025. URL: <https://www.gao.gov/products/gao-25-107569> (visited on 08/21/2026).

⁹¹GlobalSecurity.org. *Prospective Front-line Fighter (PFI)*. URL: <https://www.globalsecurity.org/military/world/russia/pfi.htm> (visited on 08/21/2026); Wikipedia. *Mikoyan MiG-29*. URL: https://en.wikipedia.org/wiki/Mikoyan_MiG-29 (visited on 08/21/2026).

⁹²Astronautix. *Yangel*. URL: <http://www.astronautix.com/y/yangel.html> (visited on 08/21/2026); HandWiki. *Biography: Vladimir Chelomey*. URL: https://handwiki.org/wiki/Biography:Vladimir_Chelomey (visited on 08/21/2026).

confined to the earliest paperwork stage of design, while a 1968 Communist Party and Council of Ministers resolution ordered the “amalgamation of small duplicative organizations” to cut the cost of exactly this kind of overlap.⁹³ Patronage mattered too: Chelomei’s bureau grew into a rival of Korolev’s largely because he had hired Khrushchev’s son as an engineer, and Khrushchev personally overrode Korolev’s proposals to keep the multi-bureau arrangement intact.⁹⁴

Post-Soviet Russia spent two decades re-consolidating a shattered industry into exactly the single points of failure the 1954 Yangel split was designed to prevent. United Aircraft Corporation formed in 2006, merging Sukhoi, MiG, Tupolev, Ilyushin, Irkut, and Yakovlev into one company, with Sukhoi and MiG fully fused as recently as 2022; United Shipbuilding Corporation followed in 2007; and Rostec, chartered the same year, today oversees roughly 800 enterprises and, by the Kremlin’s own reported account, supplies 80 percent of the materiel reaching the war in Ukraine.⁹⁵

Russia still surged drone output, and that is precisely the counterexample this section has to answer. In early 2023 the Alabuga special economic zone in Tatarstan signed a \$1.75 billion license deal with Iran to build, and in time indigenize, 6,000 Shahed-136 drones by September 2025.⁹⁶ Weekly strike volume rose from roughly 200 before September 2024 to more than 1,000 by March 2025, and in November 2025, following a Putin directive issued the previous December, Russia stood up a dedicated Unmanned Systems Forces branch to centralize drone warfare under military command, growing to roughly 114,000 personnel within six months.⁹⁷

This is capital expansion at one licensed site plus command mobilization, not decentralization. Ukraine’s output growth came from an expanding number of independent producers; Russia’s came from a single facility and a conscripted supply chain running through the same consolidated primes the 2006–07 mergers created. Mobilization multiplied an existing design under central direction, and it did not replicate the weeks-long loop between battlefield failure and the next prototype that a genuinely distributed base generates on its own. Mobilization is not deconsolidation.

⁹³Central Intelligence Agency, Directorate of Intelligence, Office of Scientific and Weapons Research. *Competition in Soviet Weapon Research and Development*. SW 88-10008X. Central Intelligence Agency, Feb. 1988. URL: https://www.cia.gov/readingroom/docs/DOC_0000500621.pdf (visited on 08/21/2026).

⁹⁴The Internet Encyclopedia of Science (David Darling). *Chelomei, Vladimir Nikolaevich*. URL: <https://www.daviddarling.info/encyclopedia/C/Chelomei.html> (visited on 08/21/2026).

⁹⁵Andrew S. Bowen. *Russian Arms Sales and Defense Industry*. R46937. Congressional Research Service, Oct. 14, 2021. URL: https://www.congress.gov/crs_external_products/R/PDF/R46937/R46937.1.pdf (visited on 08/21/2026).

⁹⁶David Albright et al. *Alabuga’s Greatly Expanded Production Rate of Shahed 136 Drones*. Institute for Science and International Security, May 10, 2024. URL: https://isis-online.org/uploads/isis-reports/documents/Alabuga%E2%80%99s_Greatly_Expanded_Production_Rate_of_Shahed_136_Drones_May_10_2024_Final.pdf (visited on 08/21/2026).

⁹⁷Benjamin Jensen and Yasir Atalan. *Drone Saturation: Russia’s Shahed Campaign*. Center for Strategic and International Studies, May 13, 2025. URL: <https://www.csis.org/analysis/drone-saturation-russias-shahed-campaign> (visited on 08/21/2026); Wikipedia. *Unmanned Systems Forces of Russia*. URL: https://en.wikipedia.org/wiki/Unmanned_Systems_Forces_of_Russia (visited on 08/21/2026).

5 What the 2025–2026 Reforms Did — and Did Not — Do

Congress and the Pentagon spent 2025 and 2026 on the largest rewrite of defense-acquisition law in a generation. Every enacted provision from that rewrite is a process lever: it changes how fast the Department of Defense (DoD) can buy from suppliers who already sell to it. Not one changes who is allowed to sell, how many suppliers must exist, or whether anyone can act on the consolidation the 1993 “Last Supper” produced. That asymmetry proves the U.S. political system can move fast on acquisition process when it chooses to — which makes the absence of structural remedy for the base itself a choice, not a constraint.

The Enacted Record

The FY26 National Defense Authorization Act (NDAA), signed December 18, 2025, authorized \$900.6 billion after the House passed it 312–112 and the Senate 77–20 — the 65th consecutive NDAA, and the legislative landing point for the House’s SPEED Act and the Senate’s FoRGED Act.⁹⁸ Section 811 repeals or amends a package of prior-year acquisition statutes; the exact count is itself instructive. Holland & Knight counts 86 provisions; Greenberg Traurig counts 75; Miller & Chevalier counts “more than 60.”⁹⁹ War on the Rocks confirms the final number was cut down from a more aggressive House and Senate draft, with dozens of additional repeals dropped in conference.¹⁰⁰ The commonly cited figure of roughly 80 repeals sits inside that 60–86 range depending on whose methodology is used — a spread that is itself evidence the number was a political marker, not a precise inventory. What none of the counts dispute is the category: every repealed provision constrains how DoD buys, and not one of them touches who is allowed to sell.

Secretary of War Pete Hegseth’s November 7, 2025 memorandum redesignated the acquisition system the “Warfighting Acquisition System” and created Portfolio Acquisition Executives (PAEs): senior officials who consolidate program management by mission and technology area, gain authority to shift funding within their own portfolio, and act as the single accountable official for a portfolio’s outcomes.¹⁰¹ The FY26 NDAA (Sections 1802–1803) codified the PAE role in statute, and all major acquisition activity is due to move under PAE portfolios within two years of the memo.¹⁰² PAEs reorganize how the department manages the programs it already

⁹⁸Holland & Knight, *FY 2026 National Defense Authorization Act*, op. cit.

⁹⁹Idem, *FY 2026 National Defense Authorization Act*, op. cit.; Greenberg Traurig, *FY 2026 NDAA: The Substantial Impact of the Fiscal Year 2026 National Defense Authorization Act on Federal Procurement Law*. Feb. 10, 2026. URL: <https://www.gtlaw.com/en/insights/2026/2/fy2026ndaa/fy-2026-ndaa-the-substantial-impact-of-the-fiscal-year-2026-national-defense-authorization-act-on-federal-procurement-law> (visited on 08/21/2026); Miller & Chevalier, *Key Provisions for Government Contractors in the 2026 NDAA*. Dec. 17, 2025. URL: <https://www.millerchevalier.com/publication/key-provisions-government-contractors-2026-ndaa> (visited on 08/21/2026).

¹⁰⁰War on the Rocks. *An Industry Guide to the FY2026 NDAA*. Dec. 23, 2025. URL: <https://warontherocks.com/2025/12/an-industry-guide-to-the-fy2026-ndaa/> (visited on 08/21/2026).

¹⁰¹Wiley Rein LLP. *Secretary Hegseth Issues Directives to Transform Acquisition Process and Defense Industrial Base*. Nov. 7, 2025. URL: <https://www.wiley.law/newsletter-Secretary-Hegseth-Issues-Directives-to-Transform-Acquisition-Process-and-Defense-Industrial-Base> (visited on 08/21/2026).

¹⁰²Holland & Knight, *FY 2026 National Defense Authorization Act*, op. cit.; Inside Government Contracts.

runs. They do not change who is eligible to bid to run them — and the one PAE-adjacent move that does touch a whole mission area, the July 2026 Direct Reporting Portfolio Manager for Unmanned Systems, explicitly excludes Major Defense Acquisition Programs from its authority, carving the biggest sole-source chokepoints back out of the one consolidation with real teeth.¹⁰³

Two further threads are demand-side, not supply-side. Executive Order (EO) 14307, “Unleashing American Drone Dominance” (June 6, 2025), directs federal agencies to expand domestic drone production and prioritize U.S.-built systems in procurement, but does not touch the structure of the manufacturing base it is trying to grow.¹⁰⁴ Hegseth’s companion July 10, 2025 memo ordered every service to field expendable small drones by the end of 2026 and reclassified small unmanned aircraft as consumable commodities rather than durable property — a genuine accounting fix, still process rather than structure.¹⁰⁵ The Army’s FY26 budget request raised small-drone procurement funding to \$726 million plus \$56 million in research funding, roughly eight times the prior year’s \$98.4 million, with a further \$581 million on the unfunded list.¹⁰⁶

The Gap

The reform wave’s own paper trail names ten policy levers. Deregulation, requirements-process speed, commercial onramps, and portfolio management are enacted. Dual-source mandates, divestiture authority, portfolio splits across suppliers, and structural chokepoint review are not.¹⁰⁷ Section 832 of the NDAA comes closest to a structural fix and still is not one: it creates panels that must decide within 14 days whether to qualify a proposed alternate supplier, but it does not require the department to seek, fund, or maintain a second source for any sole-source program.¹⁰⁸

Section 857 mergers-and-acquisitions (M&A) notification, layered on with February 2026 guidance, requires merging parties to notify the department during antitrust review. It grants no independent power to block a deal or compel divestiture, and the Federal Trade Commission and Justice Department retain sole clearance authority.¹⁰⁹ None of this touches the baseline.

From DAS to WAS: Secretary Hegseth’s Acquisition Overhaul and What It Means for Industry. Dec. 3, 2025. URL: <https://www.insidegovernmentcontracts.com/2025/12/from-das-to-was-secretary-hegseths-acquisition-overhaul-and-what-it-means-for-industry/> (visited on 08/21/2026).

¹⁰³Idem, *From DAS to WAS: Secretary Hegseth’s Acquisition Overhaul and What It Means for Industry*, op. cit.

¹⁰⁴Executive Office of the President. *Executive Order 14307: Unleashing American Drone Dominance.* Federal Register, June 6, 2025. URL: <https://www.govinfo.gov/content/pkg/FR-2025-06-11/pdf/2025-10814.pdf> (visited on 08/21/2026).

¹⁰⁵National Guard Association of the United States. *Hegseth Moves to Unleash U.S. Drone Dominance.* July 15, 2025. URL: <https://www.ngaus.org/newsroom/hegseth-moves-unleash-us-drone-dominance> (visited on 08/21/2026).

¹⁰⁶Inside Unmanned Systems. *Report: What Unmanned Systems Is America’s Military Buying in 2026?* Nov. 20, 2025. URL: <https://insideunmannedsystems.com/report-what-unmanned-systems-is-americas-military-buying-in-2026/> (visited on 08/21/2026).

¹⁰⁷Holland & Knight, *FY 2026 National Defense Authorization Act*, op. cit.

¹⁰⁸Ibid.

¹⁰⁹Greenberg Traurig. *New Guidance for Defense Sector M&A Review.* Feb. 26, 2026. URL: <https://www.gtlaw.com/en/insights/2026/2/new-guidance-for-defense-sector-m-and-a-review> (visited on 08/21/2026).

Five firms — Lockheed Martin, RTX, General Dynamics, Boeing, and Northrop Grumman — drew about one-third of all DoD contract obligations from 2019 to 2022, and by 2024 at least one of them served as prime contractor on 74 percent of major weapons systems.¹¹⁰

The one proposal with real structural teeth — the Warren-Blumenthal-Khanna bill, which would require Pentagon review of any private-equity transaction taking a 25-percent-or-larger stake in a defense contractor — was introduced in June 2026 and remained unenacted as of that August.¹¹¹ Appropriators are not simply rubber-stamping the process wave either: the January 2026 defense minibuss called DoD’s request for further budget-flexibility authority “premature” and flatly rejected the Army’s bid to raise its reprogramming-notification threshold, the mechanism its own drone-buying plan depended on.¹¹² Political appetite has not caught up to the process wave.

The Reconciliation Warning

The clearest sign that process reform has outrun demand reform is in the money itself. The Congressional Budget Office’s analysis of DoD’s FY26 budget request found that total acquisition funding rose 18 percent above the prior year’s enacted level — but base-budget acquisition funding, stripped of the one-time 2025 reconciliation act, *fell* 9.0 percent over the same period.¹¹³ Navy and Air Force acquisition each drew roughly 29 percent of their FY26 total from reconciliation money that Congress made available for obligation only through fiscal year 2029; Navy base-budget shipbuilding funding alone came in 44 percent below what the department’s own prior five-year plan anticipated, backfilled entirely by the one-time pot.¹¹⁴

This matters for the remedy, not just the ledger. Every structural fix this paper prescribes — dual-sourcing, portfolio splits, entry protection — depends on suppliers reading a stable, multiyear demand signal before they invest in a second production line. A budget that hits its topline with expiring reconciliation dollars while the base line shrinks underneath it sends the opposite signal: the same kind of demand instability that produced the 1993 consolidation this paper is arguing to reverse, arriving again through the back door of the appropriations process. Process was reformed; structure was not.

¹¹⁰American Bar Association, Public Contract Law Journal. “DoD Antitrust: Improving the Office of Industrial Base Policy’s Oversight of Mergers and Acquisitions.” In: *Public Contract Law Journal* (Oct. 9, 2025). URL: https://www.americanbar.org/groups/public_contract_law/resources/journal/2025-summer/dod-antitrust-improving-office-industrial-base-policy-oversight-mergers-acquisitions/ (visited on 08/21/2026).

¹¹¹Federal News Network. *Private Equity’s Defense Deals Could Come Under Increased Scrutiny After Midterms*. Aug. 19, 2026. URL: <https://federalnewsnetwork.com/congress/2026/08/private-equitys-defense-deals-could-come-under-increased-scrutiny-after-midterms/> (visited on 08/21/2026).

¹¹²Federal News Network. *Congress Pushes Back on Parts of DoD’s Acquisition Reform Agenda*. Jan. 20, 2026. URL: <https://federalnewsnetwork.com/congress/2026/01/congress-pushes-back-on-parts-of-dods-acquisition-reform-agenda/> (visited on 08/21/2026).

¹¹³Congressional Budget Office. *DoD’s 2026 Budget Request and Plan for Funding Provided by the 2025 Reconciliation Act*. Congressional Budget Office, Apr. 2026. URL: <https://www.cbo.gov/system/files/2026-04/62301-DoD-2026-Budget-Request.pdf> (visited on 08/21/2026).

¹¹⁴*Ibid.*

6 Recommendations

The reform wave documented above fixed how the department buys. It left untouched who is allowed to sell, and in how many pieces. Each recommendation below names an actor, a vehicle, and a price, keyed to a measured historical precedent and the condition under which that precedent held. None calls for competition everywhere: dual-sourcing where volume and learning curves clear a breakeven line, entry protection where new suppliers already qualify, divestiture only where both fail.

Recommendation 1: Mandate Dual-Sourcing Where the Breakeven Math Clears

Congress should mandate dual-sourcing by statute in chokepoint segments that clear RAND's Required Cost Reduction (RCR) breakeven threshold, not as a blanket rule across the industrial base. The evidentiary floor is the F100/F110 "Great Engine War" of 1984–1994: the Government Accountability Office's (GAO) 2009 retrospective found 30 percent cumulative acquisition savings, 16 percent operations-and-support savings, and a 94–96 percent drop in unscheduled engine removals, for a joint development cost of roughly \$610 million against a \$1.5 billion clean-sheet-engine baseline, achievable only because both competing engines were derivatives of existing designs, not new ones.¹¹⁵ A peer-reviewed 14-program panel of tactical-missile systems, selection-bias corrected, finds a comparable 20.4 percent average procurement-cost reduction from dual-sourcing.¹¹⁶ The saving is not free: GAO's own 1984 audit found a split award captured only about two-thirds of the savings available from full competition followed by a single award, a real dual-source tax the Air Force judged worth paying without ever separately quantifying the benefit.¹¹⁷ RAND's RCR formula makes the condition explicit: the median historical missile and ship program cleared only about 17 percent in realized savings and none topped 40 percent, success requires a learning-curve slope flatter than 85 percent and a production run large enough to amortize the second line.¹¹⁸ The mandate should trigger off that screen, not off the segment. Solid rocket motors (SRMs) and 155mm-class munitions, where volume is already rising and new entrants are already qualifying, clear it; low-volume, high-fixed-cost systems such as ICBMs and ballistic-missile submarines do not.

¹¹⁵Gertler. *F-35 Alternate Engine Program: Background and Issues for Congress*. Congressional Research Service, Jan. 10, 2012. URL: https://www.congress.gov/crs_external_products/R/PDF/R41131/R41131.15.pdf (visited on 08/21/2026); Camm and Glennan, *The Development of the F100-PW-220 and F110-GE-100 Engines: A Case Study of Risk Assessment and Risk Management*, op. cit.

¹¹⁶Lyon, "Does Dual Sourcing Lower Procurement Costs?" Op. cit.

¹¹⁷U.S. General Accounting Office, *Review of the Alternate Fighter Engine Competition*, op. cit.

¹¹⁸Arena and Birkler, *Determining When Competition Is a Reasonable Strategy for the Production Phase of Defense Acquisition*, op. cit.

Recommendation 2: Lock Portfolio Splits Into Statute Before the Market Reconsolidates

The Department of Defense (DoD) should split portfolio buys across suppliers by statute, not by internal reorganization alone, wherever a program’s own competitive-prototyping phase has already proved the model, and ask Congress to write the split into the authorizing statute so it outlives any single Portfolio Acquisition Executive’s tenure. The Collaborative Combat Aircraft (CCA) program is the live natural experiment: at the April 2024 Increment 1 down-select, the Air Force chose Anduril Industries and General Atomics and rejected Boeing, Lockheed Martin, and Northrop Grumman, all three of which competed and lost.¹¹⁹ Production contracts for the first 150 aircraft, awarded June 2026, target a unit cost under \$30 million.¹²⁰ Set that against the sole-sourced alternative: Northrop Grumman’s fixed-price B-21 Raider contract has booked more than \$2.1 billion in accumulated losses through the third quarter of fiscal 2025, with no competing bid ever available to check whether the government’s price is the right one.¹²¹ The FY26 NDAA’s Portfolio Acquisition Executive structure gives DoD the machinery to run CCA-style splits across more of the portfolio; it does not require doing so, and the market can reconsolidate around the primes without a statutory floor — CCA’s own Increment 2 has already reopened its vendor pool to Lockheed, Northrop, and RTX. Golden Dome’s space-based interceptor layer is the closing window to apply the lesson: reporting describes Anduril, Palantir, and SpaceX working alongside the primes on the largest acquisition line in the future defense budget, with no vendor structure locked yet.¹²² The window will not stay open.

Recommendation 3: Make Concentration a Binding Factor in Merger Review

The Federal Trade Commission (FTC) and DoD should codify defense-industrial concentration as an explicit, binding factor in merger review, and pre-clear the security-clearance mechanics that make defense divestitures uniquely hard. The precedent already exists but is not law: in January 2022 the FTC voted 4–0 to block Lockheed Martin’s \$4.4 billion acquisition of Aerojet Rocketdyne, then the country’s last independent large solid-rocket-motor maker, on a vertical-foreclosure theory, and Lockheed abandoned the deal within three weeks.¹²³ A year later the

¹¹⁹Congressional Research Service. *U.S. Air Force Collaborative Combat Aircraft*. Congressional Research Service, Nov. 28, 2025. URL: https://www.congress.gov/crs_external_products/IF/PDF/IF12740/IF12740.5.pdf (visited on 08/21/2026).

¹²⁰Air & Space Forces Magazine. *Air Force, General Atomics, Anduril Ink CCA Production Contracts*. June 2026. URL: <https://www.airandspaceforces.com/air-force-general-atomics-anduril-cca-production-contracts/> (visited on 08/21/2026).

¹²¹Northrop Grumman Corporation. *Form 10-Q: Quarterly Report for the Period Ended September 30, 2025*. U.S. Securities and Exchange Commission, Sept. 30, 2025. URL: <https://www.sec.gov/Archives/edgar/data/1133421/000113342125000053/noc-20250930.htm> (visited on 08/21/2026).

¹²²Technology.org. *Anduril and Palantir Team Up to Write the Code Behind America’s \$185 Billion Golden Dome Missile Shield*. Mar. 25, 2026. URL: <https://www.technology.org/2026/03/25/anduril-palantir-write-code-americas-185-billion-golden-dome-missile-shield/> (visited on 08/21/2026).

¹²³Federal Trade Commission. *FTC Sues to Block Lockheed Martin Corporation’s \$4.4 Billion Vertical Acquisition of Aerojet Rocketdyne*. Jan. 25, 2022. URL: <https://www.ftc.gov/news-events/news/press-releases/2022/01/ftc-sues-block-lockheed-martin-corporations-44-billion-vertical-acquisition-aerojet-rocketdyne> (visited on 08/21/2026).

FTC let L3Harris buy the same company on the strength of a voluntary, non-binding promise to keep selling motors to Lockheed’s rivals; three members of Congress publicly flagged the promise’s lack of enforceable teeth within months.¹²⁴ The same scarce national asset was ruled anticompetitive in one buyer’s hands and approved in another’s — exactly the ad hoc pattern a statute should end. DoD’s own merger-review office, meanwhile, formally assesses roughly 10 percent of the estimated 400 annual defense-related mergers with a staff of two to three people, and has not routinely evaluated national-security effects as a distinct factor.¹²⁵ Congress should require the FTC and the Justice Department to weigh industrial-base concentration as a standing factor in defense-sector review, not a discretionary add-on, and require DoD to pre-clear the Foreign Ownership, Control, or Influence (FOCI) and security-clearance approval chain for any cleared defense unit *before* a conditional merger approval is granted. A divestiture remedy, if ordered, can then close instead of stalling in a second review layer no ordinary commercial merger faces.

Recommendation 4: Fund the Entrants Already Clearing the Bar

Congress should fund entry, not just speed it, backing the solid-rocket-motor and munitions entrants already qualifying with multiyear procurement authority and government-owned technical-data packages (TDPs). The SRM industry consolidated from roughly 5,000 supplier firms twenty years ago to about 1,000 today, and DoD itself says current demand can sustain only two incumbent manufacturers.¹²⁶ Entry is happening anyway: Anduril’s Adranos subsidiary is building a \$1 billion, 500-acre SRM and drone-manufacturing complex in Ohio with production targeted for July 2026, while separately expanding its Mississippi motor line from about 600 to more than 6,000 motors a year;¹²⁷ Ursa Major matched a \$12.5 million Navy/Office of Strategic Capital investment to \$25 million developing its Lynx motor line as a Navy-sponsored second source, backed by a \$100 million private funding round;¹²⁸ and X-Bow Systems won a \$64 million Navy/Army contract in October 2023 that the Pentagon itself described as breaking the Northrop-Aerojet duopoly, because the incumbents could not meet hypersonic-program volume and price.¹²⁹ Congress should extend two mechanisms already funding entry in 155mm-shell

¹²⁴Elizabeth Warren, John Garamendi, and Mark Pocan. *Senator Warren, Representatives Garamendi, Pocan Raise Concerns, Call for Increased Oversight, Transparency from DoD Following the L3Harris-Aerojet Merger*. Aug. 2023. URL: <https://www.warren.senate.gov/oversight/letters/senator-warren-representatives-garamendi-pocan-raise-concerns-call-for-increased-oversight-transparency-from-dod-following-the-l3harris-aerojet-merger> (visited on 08/21/2026).

¹²⁵U.S. Government Accountability Office. *Defense Industrial Base: DOD Needs Better Insight into Risks from Mergers and Acquisitions*. U.S. Government Accountability Office, Oct. 17, 2023. URL: <https://www.gao.gov/products/gao-24-106129> (visited on 08/21/2026).

¹²⁶Idem, *Solid Rocket Motors: DOD and Industry Are Addressing Challenges to Minimize Supply Concerns*, op. cit.

¹²⁷Defense News. *Anduril to Build ‘Arsenal-1’ Autonomous Weapons Plant in Central Ohio*. Jan. 16, 2025. URL: <https://www.defensenews.com/industry/2025/01/16/anduril-to-build-arsenal-1-autonomous-weapons-plant-in-central-ohio/> (visited on 08/21/2026).

¹²⁸Breaking Defense. *Ursa Major Wins, Matches \$12.5M Investment from DoD to Improve Solid Rocket Motor Production*. Sept. 18, 2024. URL: <https://breakingdefense.com/2024/09/ursa-major-wins-matches-12-5m-investment-from-dod-to-improve-solid-rocket-motor-production/> (visited on 08/21/2026).

¹²⁹Jason Sherman. *DOD Qualifies New Solid-Rocket Motor Supplier, Breaks Northrop-Aerojet Domestic Duopoly*. Oct. 6, 2023. URL: <https://insidedefense.com/daily-news/dod-qualifies-new-solid-rocket-motor->

production. Multiyear procurement authority, which the FY26 NDAA extended to SM-3, THAAD, Tomahawk, LRASM, and PAC-3 on top of its FY24 extension to munitions, lets a producer commit capital against an order book longer than one contract cycle.¹³⁰ Government-owned TDP transfer let General Dynamics stand up a new Camden, Arkansas facility, for a projectile it had never produced before, to 50,000 rounds of monthly capacity.¹³¹ Fund entrants already past the technical-qualification bar, not winners picked cold; the SRM entrants above are all past it.

Recommendation 5: Move Reconciliation Funding Into the Base Budget

Congress should stabilize the demand signal by moving reconciliation-funded acquisition into the base budget, so second sources can amortize their investment against a signal that outlasts one funding cycle. The 2025 reconciliation act provided \$156 billion in one-time DoD funding available for obligation only through fiscal year 2029; the department's FY26 request leaned on it hard, with Navy and Air Force acquisition each drawing roughly 29 percent of their total from reconciliation money and Navy base-budget shipbuilding coming in 44 percent below the department's own prior plan.¹³² Total acquisition funding looks healthy, up 18 percent year over year, only because base-budget acquisition funding actually fell 9.0 percent underneath it.¹³³ Every other recommendation in this section depends on suppliers believing that signal outlasts the funding cycle. RAND's own history supplies the cautionary case: the Navy spent about \$300 million qualifying Pratt & Whitney as a second source for General Electric's F404 engine, then ordered only about 200 engines and terminated the competition, stranding the fixed cost of standing up competition against a demand signal that evaporated.¹³⁴ Congress should fold reconciliation-funded procurement lines into the base discretionary baseline before the FY2029 cliff forces another round of program cancellations. This is the 1993 lesson in reverse: consolidation followed a demand collapse the government itself engineered, and an unstable demand signal, even one masked by a temporary funding surge, produces the same incentive to consolidate rather than compete.

Sequencing

Not all five recommendations move on the same clock. Recommendations 2 and 4 need no new grant of authority. DoD can write CCA-style splits into the next tier of already multi-vendor-capable buys using authority it holds today; Congress can extend multiyear procurement

[supplier-breaks-northrop-aerojet-domestic-duopoly](#) (visited on 08/21/2026).

¹³⁰Ronald O'Rourke. *Multiyear Procurement (MYP) and Block Buy Contracting in Defense Acquisition: Background and Issues for Congress*. Congressional Research Service, Dec. 2025. URL: https://www.congress.gov/crs_external_products/R/PDF/R41909/R41909.145.pdf (visited on 08/21/2026).

¹³¹Elizabeth Urbaniak. *New Production Facility Opens to Streamline Army Artillery Production*. Aug. 11, 2025. URL: <https://www.army.mil/article/287732> (visited on 08/21/2026).

¹³²Congressional Budget Office, *DoD's 2026 Budget Request and Plan for Funding Provided by the 2025 Reconciliation Act*, op. cit.

¹³³*Ibid.*

¹³⁴Birkler, Dews, and Large, *Issues Associated with Second-Source Procurement Decisions*, op. cit.

and technical-data-package funding through the FY27 NDAA and appropriations cycle using mechanisms it has already used for 155mm shells. Recommendation 1’s dual-source mandate is a medium lift — it requires Congress to legislate the RCR screen itself — but it can start narrow, SRMs first, where the case is strongest and entrants already exist, inside the FY27 NDAA. Recommendations 3 and 5 are the harder, slower levers. Codifying merger review means reconciling FTC, Justice Department, and DoD authority in a way the pending Warren-Blumenthal-Khanna bill has not yet achieved; moving reconciliation-funded acquisition into the base budget requires the exact appropriations-structure change that FY26 appropriators already called “premature.” Both are FY28-and-beyond projects. The FY27 cycle decides whether the process-versus-structure choice this paper documents starts closing, or hardens into the next decade’s baseline.

7 Objections

Five objections recur in early readings of this argument. Each is stated at full strength, then answered or conceded within a defined scope.

“Consolidation Never Raised Prices”

The strongest empirical challenge to this paper is econometric. Rodrigo Carril and Mark Duggan’s peer-reviewed, causally identified study of 1990s merger effects on Department of Defense (DoD) contracting found the top five contractors’ share of defense dollars rose from 21.7 percent in 1990 to 31.3 percent in 2000, that consolidation reduced competitive, multi-bid awards, and pushed contracts from fixed-price toward cost-plus, with no significant rise in acquisition costs: DoD’s monopsony power offset whatever pricing power sellers gained.¹³⁵ The price-gouging version of a deconsolidation thesis does not survive that finding.

This paper’s case does not rest on price. It rests on chokepoint concentration, surge failure, and iteration speed: by October 2025 the Army’s 155mm shell line, expanded since 2022, still ran short of its 100,000-round-a-month target, now slipped to mid-2026.¹³⁶ History credits competition with speed, not savings: a 2012 Congressional Research Service retrospective on the Great Engine War found its primary benefit was contractor responsiveness, not dollar savings.¹³⁷

¹³⁵Rodrigo Carril and Mark Duggan. “The Impact of Industry Consolidation on Government Procurement: Evidence from Department of Defense Contracting.” In: *Journal of Public Economics* 184 (2020). URL: <https://rcarril.github.io/website/mergersJPUBEaccepted.pdf> (visited on 08/21/2026).

¹³⁶Breaking Defense, “Army Hitting Stride with 155mm Production, but Key General Worries over What’s Needed Next,” op. cit.

¹³⁷Jeremiah Gertler. *F-35 Alternate Engine Program: Background and Issues for Congress*. R41131. Congressional Research Service, Jan. 10, 2012. URL: https://www.congress.gov/crs_external_products/R/PDF/R41131/R41131.15.pdf (visited on 08/21/2026).

“Monopsony Makes Breakup Futile”

The structural rebuttal is harder to wave off. Jacques Gansler, DoD’s own acquisition chief from 1997 to 2001, argued the department should be modeled as a monopsony firm whose demand behavior, not supplier market power, sets the industry’s incentives.¹³⁸ Margaret Mullins’s 2025 retrospective quotes him calling the defense sector a single buyer with one or two suppliers of any given weapon system.¹³⁹ Splitting suppliers without touching DoD’s cost-plus drift and single-bid tolerance recreates the conditions that produced the 1993 Last Supper.

That correctly diagnoses an incomplete remedy; it does not rebut this one. Recommendation 5 pairs market structure with demand-signal reform rather than antitrust or divestiture alone, because a supply-side fix without a demand-side fix fails for exactly the reason Gansler identifies.

“Duplication Costs Too Much”

History backs this objection more than dual-sourcing’s defenders admit. The Government Accountability Office’s (GAO) 1984 review of the Great Engine War found a single-winner award would have saved 15 percent of life-cycle cost against 10 percent for the actual 75/25 split, a 5-percentage-point dual-source tax the Air Force judged worth paying but never quantified.¹⁴⁰ The Great Engine War is the outlier: a 1987 Office of the Secretary of Defense (OSD) study of the Sparrow and Sidewinder missile lines found dual-sourcing cost 11 to 42 percent more than sole-sourcing in 11 of 12 tests,¹⁴¹ and GAO’s 1984 government-wide audit found only 1 of 9 buying offices had studied cost-benefit at all — a \$600,000-to-\$4.6-million loss.¹⁴²

That history is why this paper’s remedy is conditional, not universal. RAND’s Required Cost Reduction formula gives a testable breakeven line: dual-sourcing clears only where the learning-curve slope is flat enough and the production run large enough to amortize entry cost, empirically 17 percent required savings for missiles and ships and 30 percent for electronics; below that line the formula returns impossible.¹⁴³ Segments that clear the threshold get mandated dual-sourcing; segments that fall short get entry support instead.

¹³⁸John D. Driessnack. *Defense Market(s): A Relook at the Explanatory Power of Several Economic Schools of Thought while Viewing the Defense Department as a Monopsony Firm*. SYM-AM-25-362. Naval Postgraduate School, Acquisition Research Program, May 5, 2025. URL: <https://dair.nps.edu/bitstream/123456789/5414/1/SYM-AM-25-362.pdf> (visited on 08/21/2026).

¹³⁹Margaret Mullins. *The Myths of the Last Supper: The Lessons of History and the Future of Defense Procurement*. Vanderbilt Policy Accelerator, Sept. 2025. URL: <https://cdn.vanderbilt.edu/vu-URL/wp-content/uploads/sites/412/2025/09/11175053/The-Myths-of-the-Last-Supper.pdf> (visited on 08/21/2026).

¹⁴⁰U.S. General Accounting Office, *Review of the Alternate Fighter Engine Competition*, op. cit.

¹⁴¹U.S. Office of the Secretary of Defense, Program Analysis and Evaluation. *Dual Sourcing in Defense Missile Procurement*. U.S. Office of the Secretary of Defense, Apr. 8, 1987. URL: https://economics.osd.mil/dual_sourcing.pdf (visited on 08/21/2026).

¹⁴²U.S. General Accounting Office. *Cost Effectiveness of Dual Sourcing for Production Price Competition Is Uncertain*. NSIAD-84-111. U.S. General Accounting Office, Aug. 31, 1984. URL: <https://www.gao.gov/assets/nsiad-84-111.pdf> (visited on 08/21/2026).

¹⁴³Arena and Birkler, *Determining When Competition Is a Reasonable Strategy for the Production Phase of Defense Acquisition*, op. cit.

“New Entrants Already Fix This”

Solid rocket motors, the tightest chokepoint the Federal Trade Commission (FTC) and GAO have flagged, are seeing real new entry. Anduril acquired Adranos in 2023 and is building a \$1 billion, 5-million-square-foot Arsenal-1 factory in Ohio for July 2026 production.¹⁴⁴ Ursa Major is developing a Navy-sponsored second-source motor on a \$25 million joint DoD investment and has raised a \$100 million Series E.¹⁴⁵ X-Bow won a \$64 million Navy and Army contract the Pentagon credited with breaking the Northrop-Aerojet duopoly.¹⁴⁶ If the market is already fixing this, a mandate solves an already-solved problem.

Entry is real but unproven and fragile. General Dynamics’ \$469 million Mesquite, Texas plant, the diversification wave’s largest piece, had produced zero contracted parts two years in, per the Pentagon’s inspector general.¹⁴⁷ Nor does the base rate favor unprotected entrants: this market went from 13 suppliers to 3 in tactical missiles and 8 to 3 in fixed-wing aircraft within a decade of the Last Supper, with no entry protection, and no entrant today has survived the down-cycle that drove it.¹⁴⁸ Entry earns support, not a pass on structural protection.

“The Best Academic Case for Consolidation”

Eugene Gholz and Harvey Sapolsky’s peer-reviewed study of the 1990s drawdown is the strongest scholarly case against this paper’s remedy. They argued supply-side production competition is a weak lever under monopsony: DoD’s contract-auditing apparatus, not seller count, disciplines cost, and teaming or subcontracting can preserve design competition without paying to duplicate production lines. “The empirical record suggests that this benefit, too, is exaggerated,” they wrote, crediting interservice rivalry, not seller count, as the real innovation engine.¹⁴⁹

The concession is bounded: Gholz and Sapolsky correctly diagnosed 1999’s defense sector, an overcapacity glut, a monopsony buyer with real audit leverage, interservice rivalry still generating new designs. None of that describes the 2020s chokepoints this paper maps. Teaming preserves a design team on paper, not a second production line: the Army’s 155mm line, expanded since 2022, still ran short of target.¹⁵⁰ And the buyer-side discipline they trusted to substitute for supplier competition is the same DoD whose own 2022 accounting shows the concentration they

¹⁴⁴Anduril Industries. *Anduril Building Arsenal-1 Hyperscale Manufacturing Facility in Ohio*. Jan. 16, 2025. URL: <https://www.anduril.com/news/anduril-building-arsenal-1-hyperscale-manufacturing-facility-in-ohio> (visited on 08/21/2026).

¹⁴⁵PR Newswire. *Ursa Major and U.S. Navy Make \$25 Million Joint Investment in New Solid Rocket Motor Prototype*. Sept. 17, 2024. URL: <https://www.prnewswire.com/news-releases/ursa-major-and-us-navy-make-25-million-joint-investment-in-new-solid-rocket-motor-prototype-302249674.html> (visited on 08/21/2026).

¹⁴⁶Sherman, *DOD Qualifies New Solid-Rocket Motor Supplier, Breaks Northrop-Aerojet Domestic Duopoly*, op. cit.

¹⁴⁷ProPublica, *General Dynamics Artillery Factory Failed*, op. cit.

¹⁴⁸U.S. Department of Defense, *State of Competition within the Defense Industrial Base*, op. cit.

¹⁴⁹Eugene Gholz and Harvey M. Sapolsky. “Restructuring the U.S. Defense Industry.” In: *International Security* 24.3 (Winter 1999/2000 1999), pp. 5–51. URL: https://www.belfercenter.org/sites/default/files/pantheon_files/files/publication/gholz_sapolsky_v24n3.pdf (visited on 08/21/2026).

¹⁵⁰Breaking Defense, “Army Hitting Stride with 155mm Production, but Key General Worries over What’s Needed Next,” op. cit.

treated as costless exit: tactical missiles from 13 suppliers to 3, fixed-wing aircraft from 8 to 3, within a decade of the drawdown they were describing.¹⁵¹

8 Conclusion

The FY27 National Defense Authorization Act (NDAA) and the accompanying Planning, Programming, Budgeting, and Execution (PPBE) reform package are the next vehicle for this argument, and a test of whether Washington’s appetite for acquisition reform extends to market structure, not just process.

Three of the five recommendations above are realistically FY27-actionable: entry funding for solid-rocket-motor chokepoints through the Defense Production Act Title III and appropriations mechanisms needs no new statute (Recommendation 4); a portfolio-split-by-statute rider extending the two-vendor Collaborative Combat Aircraft model into the NDAA is plausible (Recommendation 2); and mandating dual-sourcing where the Required Cost Reduction threshold clears, starting with solid rocket motors, is a heavier but realistic lift (Recommendation 1). The remaining two will not move as fast: merger-review codification rides on the pending Warren-Blumenthal-Khanna bill and waits for the next Congress (Recommendation 3), and demand-signal reform is the heaviest lift of all — appropriators called a similar request premature in January 2026 (Recommendation 5), the piece the monopsony objection says cannot be skipped. Deferring it guarantees the other recommendations hit the same wall Gansler predicted.

The cost of getting this wrong shows up in the Pentagon’s wargames: across roughly two dozen Taiwan Strait iterations, the United States has exhausted its entire long-range anti-ship missile inventory in the first week of combat, against the 88 missiles the FY2023 budget proposed buying that year.¹⁵² Another decade of process-only reform leaves that magazine exactly as empty as it is today.

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¹⁵¹U.S. Department of Defense, *State of Competition within the Defense Industrial Base*, op. cit.

¹⁵²Jones et al., *Empty Bins in a Wartime Environment: The Challenge to the U.S. Defense Industrial Base*, op. cit.

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